

x x x model(Heisenberg model)

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Chapter 1

XXX Model

1.1 Introduction

Hamiltonian of the model

$$H = -\frac{1}{2}J \sum_{j=1}^N \sigma_j \cdot \sigma_{j+1} - \frac{1}{2}h \sum_{j=1}^N \sigma_j^z \quad (1.1)$$

We let $h = 0$.

With definition $S^\pm = S^x \pm iS^y, S^x = \frac{1}{2}\sigma^x \dots$. We have

$$[S_j^\pm, S_i^\pm] = 0 \quad [S_i^\pm, S_j^\mp] = \pm 2S_i^z \delta_{ij} \quad (1.2)$$

$$[S_i^z, S_j^\pm] = \pm S_i^\pm \delta_{ij} \quad [S_i^\pm, S_j^z] = \mp S_i^\pm \delta_{ij} \quad (1.3)$$

We may let $h = 0$. We can then rewrite (1.1)

$$H = -\frac{1}{2}J \sum_{j=1}^N (S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+ + 2S_j^z S_{j+1}^z) - h \sum_{j=1}^N S_j^z \quad (1.4)$$

We have(can be easily verified)

$$S_j^+ |\uparrow\rangle_j = 0 \quad S_j^- |\uparrow\rangle_j = |\downarrow\rangle_j \quad S_j^z |\uparrow\rangle_j = \frac{1}{2} |\uparrow\rangle_j \quad (1.5)$$

Define vaccum state

$$|0\rangle = |\uparrow\rangle_1 \otimes |\uparrow\rangle_2 \otimes \dots \otimes |\uparrow\rangle_N \quad (1.6)$$

We can see $|0\rangle$ is the eigenstate of (1.4)

$$H |0\rangle = \left(-\frac{J}{4}N - \frac{1}{2}Nh\right) |0\rangle \equiv E_0 |0\rangle \quad (1.7)$$

We will take advantage of symmetries.

- translational invariance $\rightarrow$ superpositions of plane waves.

- Heisenberg chain possesses full $SU(2)$ rotational invariance.

One way to understand $SU(2)$ symmetry.

The generators of $SU(2)$ group are S_x, S_y, S_z or S^+, S^-, S_z which combined with the commutation relations form an algebra. Besides, $[H, \sum_i S_i^{x,y,z}] = 0$ should be satisfied to maintain $SU(2)$ invariance

We define the z -component of the total spin $S^z = \sum_{n=1}^N S_n^z$. The conservation of S^z can be easily derived by verifying that $[H, S^z] = 0$. Consequently, we can consider separately sectors defined by the quantum number $S^z = \frac{N}{2} - R$, where R is the number of down spins.

We define the state with the n th spin down as:

$$|n\rangle = S_n^- |0\rangle \quad n = 1, \dots, N \quad (1.8)$$

However $|n\rangle$ is not the eigenstate of H . We define

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{ikn} |n\rangle \quad k = 2\pi m/N, m = 0, \dots, N-1 \quad (1.9)$$

as a linear combination of $|n\rangle$, with translational symmetry (i.e., the invariance of H with respect to discrete translations) implicitly included by a phase factor.

The vectors $|\psi\rangle$ are eigenstates of translation operator with eigenvalues e^{ik} and also of H with eigenvalues

$$E = E_0 + J(1 - \cos k) \quad (1.10)$$

Question: About the translation operator and its eigenvalues e^{ik} .

Vectors (1.9) represent *magnon excitations* ($\Delta S = 1$ excitations). **The $k = 0$ state is degenerate with $|0\rangle$.** Why is this stressed.

1.2 Two-body problem

For $R = 2$, we write a generic eigenstates as

$$|\psi\rangle = \sum_{1 \leq n_1 \leq n_2 \leq N} f(n_1, n_2) |n_1, n_2\rangle \quad (1.11)$$

where $|n_1, n_2\rangle = S_{n_1}^- S_{n_2}^- |F\rangle$. We then consider the eigenvalue equation

$$H |\psi\rangle = E |\psi\rangle \quad (1.12)$$

which indicates

$$2[E - E_0]f(n_1, n_2) = J(4f(n_1, n_2) - f(n_1 - 1, n_2) - f(n_1 + 1, n_2) - f(n_1, n_2 - 1) - f(n_1, n_2 + 1)) \quad \text{for } n_2 > n_1 + 1 \quad (1.13)$$

$$2[E - E_0]f(n_1, n_2) = J(2f(n_1, n_2) - f(n_1 - 1, n_2) - f(n_1, n_2 + 1)), \quad \text{for } n_2 = n_1 + 1 \quad (1.14)$$

Bethe's preliminary ansatz for $f(n_1, n_2)$ is:

$$f(n_1, n_2) = A e^{i(k_1 n_1 + k_2 n_2)} + A' e^{i(k_1 n_2 + k_2 n_1)} \quad (1.15)$$

which automatically satisfy (1.14) with energy

$$E = E_0 + J \sum_{j=1,2} (1 - \cos k_j) \quad (1.16)$$

We have conditions for coefficients,

$$\frac{A}{A'} \equiv e^{i\theta} = -\frac{e^{i(k_1+k_2)} + 1 - 2e^{ik_1}}{e^{i(k_1+k_2)} + 1 - 2e^{ik_2}} \quad (1.17)$$

which can be rewritten as

$$2\cot \frac{\theta}{2} = \cot \frac{k_1}{2} - \cot \frac{k_2}{2} \quad (1.18)$$

The quasi-momenta k_1, k_2 of the Bethe Ansatz is determined by the periodic boundary conditions: $f(n_1, n_2) = f(n_2, n_1 + N)$ (We interchange the position of n_1 and n_2) :

$$e^{ik_1 N} = e^{i\theta}, \quad e^{ik_2 N} = e^{-i\theta} \quad (1.19)$$

Equivalently

$$Nk_1 = 2\pi I_1 + \theta, \quad Nk_2 = 2\pi I_2 - \theta \quad (1.20)$$

where the $I_j \in \{0, 1, \dots, N-1\}$ are integer quantum numbers.

We can see

$$k_1 = \frac{2\pi I_1}{N} + \frac{\theta}{N}, \quad k_2 = \frac{2\pi I_2}{N} - \frac{\theta}{N} \quad (1.21)$$

The restriction on I_j maintains k in the first Brillouin zone. The total momentum of the state is

$$K = k_1 + k_2 = \frac{2\pi}{N}(I_1 + I_2) \quad (1.22)$$

Question: Can the phase shift θ be observed. What's the physical significance. The Berry phase is a geometry structure then what about this phase. Is the total momentum conserved. Is (I_1, I_2) describing energy level.

If we restrict I_1, I_2 to $0 \leq I_1 \leq I_2 \leq N-1$, there are $\frac{N(N+1)}{2}$ pairs meet the ordering restriction. But only $\frac{N(N-1)}{2}$ satisfy (1.18) (1.20).

We get (1.23) by combining (1.18) (1.20) (1.22):

$$2\cot \frac{Nk_1}{2} = \cot \frac{k_1}{2} - \cot \frac{K - k_1}{2} \quad (1.23)$$

Let

$$k_1 \equiv \frac{K}{2} + ik, \quad k_2 \equiv \frac{K}{2} - ik \quad (1.24)$$

which, substituted into (1.20) yields

$$\theta = \pi(I_2 - I_1) + iNk \quad (1.25)$$

Equation (1.18) becomes

$$\cos \frac{K}{2} \sinh(Nk) = \sinh[(N-1)k] + \cos[\pi(I_1 - I_2)] \sinh k \quad (1.26)$$

1.3 The Bethe Solution

We expand the states with generic R overturned spins into the natural (computational) basis

$$|\psi\rangle = \sum_{1 \leq n_1 \leq \dots \leq n_R \leq N} f(n_1, \dots, n_R) |n_1, \dots, n_R\rangle \quad (1.27)$$

where

$$f(n_1, \dots, n_R) = \sum_P \exp \left(i \sum_{j=1}^R k_{P_j} n_j + \frac{i}{2} \sum_{l < j} \theta(k_{P_l}, k_{P_j}) \right) \quad (1.28)$$

where P denotes permutations and we introduce the antisymmetric phase shift $\theta(k_l, k_j) = -\theta(k_j, k_l)$. Similarly, we can extract the consistency equations for the coefficients $f(n_1, \dots, n_R)$ from eigenvalue equation $H|\psi\rangle = E|\psi\rangle$. The energy eigenvalue equation becomes

$$E = E_0 + J \sum_{j=1}^R (1 - \cos k_j) \quad (1.29)$$

and the eigenstate condition can be written as

$$2f(n_1, \dots, n_j, n_j + 1, \dots, n_R) = f(n_1, \dots, n_j, n_j, \dots, n_R) + f(n_1, \dots, n_j + 1, n_j + 1, \dots, n_R) \quad (1.30)$$

Similarly we have phase condition:

$$e^{i\theta(k_j, k_l)} = - \frac{e^{i(k_j + k_l)} + 1 - 2e^{ik_j}}{e^{i(k_j + k_l)} + 1 - 2e^{ik_l}} \quad (1.31)$$

which can be cast in real form as

$$2 \cot \frac{\theta(k_j, k_l)}{2} = \cot \frac{k_j}{2} - \cot \frac{k_l}{2}, \quad j, l = 1, \dots, R \quad (1.32)$$

Similarly the periodicity of the chain for translations by N sites $f(n_1, \dots, n_R) = f(n_2, \dots, n_R, n_1 + N)$ gives (after taking their logarithm)

$$\sum_{j=1}^R k_{P_j} n_j + \frac{1}{2} \sum_{l < j} \theta(k_{P_l}, k_{P_j}) = \frac{1}{2} \sum_{l < j} \theta(k_{P'_l}, k_{P'_j}) - 2\pi \tilde{I}_{P'R} + \sum_{j=2}^R k_{P'_{j-1}} n_j + k_{P'_R} (n_1 + N) \quad (1.33)$$

where $P'(j-1) = Pj, j = 2, \dots, R$; $P'R = P1$. By canceling the same terms on both sides of (1.33), we derive the relations between the phase shifts and the quasi-momenta (or Bethe equations):

$$Nk_j = 2\pi \tilde{I}_j + \sum_{l \neq j} \theta(k_j, k_l), \quad j = 1, \dots, R \quad (1.34)$$

We introduce the *rapidities* λ_j to parametrize the quasi-momenta:

$$\cot \frac{k_j}{2} = \lambda_j, \quad \text{or} \quad k_j = \frac{1}{i} \ln \frac{\lambda_j + i}{\lambda_j - i} = \pi - \theta_1(\lambda_j) \quad \text{or} \quad e^{ik_j} = \frac{\lambda_j + i}{\lambda_j - i} \quad (1.35)$$

where

$$\theta_n(\lambda) \equiv 2 \arctan \frac{\lambda}{n} \quad (1.36)$$

The (bare) energy and momentum of an individual magnon, characterized by a quasi-momentum k , is

$$p_0(\lambda) = \frac{1}{i} \ln \frac{\lambda + i}{\lambda - i} = k \quad (1.37)$$

$$\epsilon_0(\lambda) = -J \frac{dk}{d\lambda} = \frac{2J}{\lambda^2 + 1} = J(1 - \cos k) \quad (1.38)$$

In terms of these rapidities (1.31) (1.35) (1.36), the scattering phase is

$$\theta(k_j, k_l) = -\theta_2(\lambda_j - \lambda_l) + \pi \operatorname{sgn}[R(\lambda_j - \lambda_l)] \quad (1.39)$$

where $R(x)$ is the real part of x . The Bethe equations (1.34) become

$$N\theta_1(\lambda_j) = 2\pi I_j + \sum_{l=1}^R \theta_2(\lambda_j - \lambda_l), \quad j = 1, \dots, R \quad (1.40)$$

Equivalently

$$\left(\frac{\lambda_j - i}{\lambda_j + i} \right)^N = - \prod_{l=1}^R \frac{\lambda_j - \lambda_l - 2i}{\lambda_j - \lambda_l + 2i} \quad (1.41)$$

The state is now defined by these "new" Bethe numbers $\{I_j\}, j = 1, \dots, R$. Because (1.40) is translational invariant ($[\lambda_j - \lambda_l]$ indicates this), **two equal I_j produce the same rapidities and thus a non valid solution.**

Using the rapidities to parametrize the eigenstates, their energies and momenta are

$$E = E_0 + \sum_{j=1}^R \epsilon_0(\lambda_j) = E_0 + \sum_{j=1}^R \frac{2J}{\lambda_j^2 + 1} \quad (1.42)$$

$$K = \left(\sum_{j=1}^R p_0(\lambda_j) \right) \bmod 2\pi = \left(\pi R - \frac{2\pi}{N} \sum_{j=1}^R I_j \right) \bmod 2\pi \quad (1.43)$$

1.4 String Solutions

In this section, we take the thermodynamic limit $N \rightarrow \infty$ known as *string hypothesis*.

We first look at the $R = 2$ case. The Bethe Equation written in terms of the rapidities are:

$$\left(\frac{\lambda_1 + i}{\lambda_1 - i} \right)^N = \frac{\lambda_1 - \lambda_2 + 2i}{\lambda_1 - \lambda_2 - 2i} \quad (1.44)$$

$$\left(\frac{\lambda_2 + i}{\lambda_2 - i} \right)^N = \frac{\lambda_2 - \lambda_1 + 2i}{\lambda_2 - \lambda_1 - 2i} \quad (1.45)$$

In the limit $N \rightarrow \infty$, the LHS is strictly zero or infinity supposing $R(\lambda) \neq 0$. We thus have:

$$\lambda_1 - \lambda_2 = \pm 2i, \quad i.e. \quad \lambda \pm i \quad (1.46)$$

The energy and momentum of this state are real:

$$p_{\frac{1}{2}}(\lambda) = p_0(\lambda + i) + p_0(\lambda - i) = \frac{1}{i} \ln \frac{\lambda + 2i}{\lambda - 2i} \quad (1.47)$$

$$\epsilon_{\frac{1}{2}}(\lambda) = \epsilon_0(\lambda + i) + \epsilon_0(\lambda - i) = \frac{4J}{\lambda^2 + 4} \quad (1.48)$$

which gives the dispersion relation

$$\epsilon_{\frac{1}{2}}(p) = \frac{J}{2}(1 - \cos p_{\frac{1}{2}}) \quad (1.49)$$

For $R > 2$, we assume that complex solutions can be organized into *complexes(or strings)* of $2M + 1$ rapidities. Here $M = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ and the rapidities have the structure

$$\lambda_m^{(M)} = \lambda_M + 2im, \quad \lambda_M \text{ is real, } m = -M, -M + 1, \dots, M - 1, M \quad (1.50)$$

We denote the number of complexes of length M by ν_M . Then **a state with a given magnetization** satisfies

$$R = \sum_M (2M + 1)\nu_M \quad (1.51)$$

Since the string containing $2M + 1$ spins move together with the same real rapidity, we will treat them as a single entity. **Note: complex solutions to the Bethe equations always appear in conjugated pairs: $\{\lambda_j\} = \{\lambda_j^*\}$**

The energy and momentum of a M -complex are obtained by summing over all rapidities within one string

$$P_M(\lambda_M) = \sum_{m=-M}^M p_0(\lambda_M + 2im) = \frac{1}{i} \ln \frac{\lambda_M + i(2M + 1)}{\lambda_M - i(2M + 1)} = \pi - \theta_{2M+1}(\lambda_M), \quad (1.52)$$

$$\epsilon_M(\lambda_M) = \sum_{m=-M}^M \epsilon_0(\lambda_M + 2im) = \frac{2J(2M + 1)}{\lambda_M^2 + (2M + 1)^2} = \frac{J}{2M + 1}(1 - \cos P_M) \quad (1.53)$$

The scattering phase of a M -complex with a simple magnon(0-complex) is:

$$S_{0,M}(\lambda_0 - \lambda_M) = S_{0,M}(\lambda) = \frac{\lambda + i2M}{\lambda - i2M} \frac{\lambda + i2(M + 1)}{\lambda - i2(M + 1)} \quad (1.54)$$

and the scattering of two complexes of length M and M' is

$$S_{M,M'}(\lambda) = \prod_{L=|M-M'|}^{M+M'} S_{0,L}(\lambda) \quad (1.55)$$

We denote the real part of the j -th M type complex by $\lambda_{M,j}$ where $j = 1, \dots, \nu_M$. The Bethe equations for the complexes is obtained by grouping all the rapidities $\lambda_j^{(M)}$ belonging to the same complex and performing first the products within each complex so to be left only with consistency conditions on their real centers $\lambda_{M,j}$:

$$e^{iP_M(\lambda_{M,j})N} = \prod_{M' \ j', (M', j') \neq (M, j)} \prod_{j=1}^{\nu_{M'}} S_{M,M'}(\lambda_{M,j} - \lambda_{M',j'}), \quad \forall M; j = 1, \dots, \nu_M \quad (1.56)$$

Similarly, we take the logarithm of (1.56) and use the familiar identity

$$\frac{1}{i} \ln \frac{\lambda + in}{\lambda - in} = \pi - 2 \arctan \frac{\lambda}{n} = \pi - \theta_n(\lambda) \quad (1.57)$$

we get

$$N \theta_{2M+1}(\lambda_{M,j}) = 2\pi I_{M,j} + \sum_{(M',j') \neq (M,j)} \theta_{M,M'}(\lambda_{M,j} - \lambda_{M',j'}) \quad (1.58)$$

where

$$\theta_{M,M'}(\lambda) \equiv \sum_{L=|M-M'|}^{M+M'} [\theta_{2L}(\lambda) + \theta_{2L+2}(\lambda)] \quad (1.59)$$

and the $L = 0$ is omitted to be omitted.

Some constraint on **Bethe numbers** $\{I_{M,j}\}$:

- $I_{M,j} \neq I_{M,j'}$ in order to have a non-vanishing solution.
- Since momenta are constrained within Brillouin zone, the Bethe numbers are bounded. The bounded number is given by taking $\lambda_M^{(\infty)} = \infty$.

$$I_M^{(\infty)} = - \sum_{M' \neq M} [2 \min(M, M') + 1] \nu_{M'} - \left(2M + \frac{1}{2}\right) (\nu_M - 1) + \frac{N}{2} \quad (1.60)$$

The maximum quantum number is given by adding a M-complex shift to $I_M^{(\infty)}$:

$$I_M^{max} = I_M^{(\infty)} - (2M + \frac{1}{2}) - \frac{1}{2} = \frac{N-1}{2} - \sum_{M'} J(M, M') \nu_{M'} \quad (1.61)$$

where

$$J(M, M') \equiv \begin{cases} 2 \min(M, M') + 1 & M \neq M' \\ 2M + \frac{1}{2} & M = M' \end{cases} \quad (1.62)$$

The $\frac{1}{2}$ in (1.61) is from the $\frac{N}{2}$ in (1.60) or by taking into account that with each rapidity the Bethe numbers shift from integers to half-integers and vice-versa.

Since all the scattering phases are odd functions, we have that

$$I_{M,min} = -I_{M,max} \quad (1.63)$$

which means that there are

$$P_M = 2I_M^{max} + 1 = N - 2 \sum_{M'} J(M, M') \nu_{M'} \quad (1.64)$$

vacancies for a M-complex.

1.5 The Anti-ferromagnetic Case: $J = -1$

We consider first single quasi-particle Excitations

$$\nu_0 = \frac{N}{2} : \quad \nu_M = 0, M \geq \frac{1}{2}, \quad \rightarrow \quad R = \frac{N}{2} \quad (1.65)$$

The number of vacancies for this configuration according to (1.64) is

$$P_0 = N - 2J(0, 0)\nu_0 = N - \frac{N}{2} = \frac{N}{2} \quad (1.66)$$

which equals the number of particle states. Hence, the quantum numbers occupy all the allowed vacancies:

$$-\frac{N}{4} + \frac{1}{2} \leq I_{0,k} \leq \frac{N}{4} - \frac{1}{2} \quad (1.67)$$

Excited states over this ground state are constructed by taking away quasi-particles from the single state and moving them into complexes, we will characterize the excited states by κ , with

$$\nu_0 = \frac{N}{2} - \kappa \quad (1.68)$$

We analyse ground state again.

The linear integral equation for the density $\rho_0(\lambda)$ is

$$\rho_0(\lambda) = \frac{1}{\pi} \frac{1}{1 + \lambda^2} - \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{2}{(\lambda - \mu)^2 + 4} \rho_0(\mu) d\mu \quad (1.69)$$

which can be rewritten as

$$\rho_0(\lambda) + \frac{1}{2\pi} \int_{-\infty}^{\infty} K(\lambda - \mu) \rho_0(\mu) d\mu = \frac{1}{2\pi} \theta'_1(\lambda) \quad (1.70)$$

where we define the kernel

$$K(\lambda) \equiv \frac{d}{d\lambda} \theta_2(\lambda) = \frac{4}{\lambda^2 + 4} \quad (1.71)$$

Using Fourier transformation, we get $\rho_0(\lambda)$ as in equation (1.97)

$$\rho_0(\lambda) = \frac{1}{4 \cosh(\frac{\pi\lambda}{2})} \quad (1.72)$$

The momentum and energy of the ground state are then given by

$$K = N \int p_0(\lambda) \rho_0(\lambda) d\lambda = \frac{\pi}{2} N \bmod 2\pi \equiv K_{AFM} \quad (1.73)$$

$$E = E_0 + N \int \epsilon_0(\lambda) \rho_0(\lambda) d\lambda = N \left(\frac{1}{4} - \ln 2 \right) \equiv E_{AFM} \quad (1.74)$$

We now look at states with $\nu_0 = \frac{N}{2} - 1$ and $\nu_M = 0$ for $M \geq \frac{1}{2}$. They are characterized by two holes.(spinon excitations as in section 6) We let the empty quantum numbers are j_1 and j_2 :

$$I_{0,j} = j + \theta_H(j - j_1) + \theta_H(j - j_2) \quad (1.75)$$

where $\theta_H(x)$ is Heqviside step-function. The integral equation for the real roots rapidity density $\rho_t(\lambda)$ (where t stands for triplet) is

$$\rho_t(\lambda) + \frac{1}{2\pi} \int_{-\infty}^{\infty} K(\lambda - \mu) \rho_t(\mu) d\mu = \frac{1}{\pi} \frac{1}{1 + \lambda^2} - \frac{1}{N} [\delta(\lambda - \lambda_1) + \delta(\lambda - \lambda_2)] \quad (1.76)$$

where $\lambda_i \rightarrow x_i = \frac{j_i}{N}$. Because of the linearity of equation (1.76). We rewrite $\rho_t(\lambda)$ as:

$$\rho_t(\lambda) = \rho_0(\lambda) + \frac{1}{N} [\tau(\lambda - \lambda_1) + \tau(\lambda - \lambda_2)] \quad (1.77)$$

where $\tau(\lambda)$ is the solution of equation:

$$\tau(\lambda) + \frac{1}{2\pi} \int_{-\infty}^{\infty} K(\lambda - \mu) \tau(\mu) d\mu = -\delta(\lambda) \quad (1.78)$$

We give the momentum and energy of the state

$$K = N \int p_0(\lambda) \rho_t(\lambda) d\lambda = K_{AFM} + k(\lambda_1) + k(\lambda_2) \quad (1.79)$$

$$E = N \int \epsilon_0(\lambda) \rho_t(\lambda) d\lambda = E_{AFM} + \epsilon(\lambda_1) + \epsilon(\lambda_2) \quad (1.80)$$

The dispersion relation is

$$\epsilon(k) = \frac{\pi}{2} \sin k, \quad -\frac{\pi}{2} \leq k \leq \frac{\pi}{2} \quad (1.81)$$

We now look at the state with $\nu_0 = \frac{N}{2} - 2$, $\nu_{\frac{1}{2}} = 1$ and $\nu_M = 0$ for $M \geq 1$. For the density of real roots $\rho_s(\lambda)$ (s is for singlet) we get the integral equation

$$\rho_s(\lambda) + \frac{1}{2\pi} \int_{-\infty}^{\infty} K(\lambda - \mu) \rho_s(\mu) d\mu = \frac{1}{\pi} \frac{1}{1 + \lambda^2} - \frac{1}{N} \left(\delta(\lambda - \lambda_1) + \delta(\lambda - \lambda_2) + \frac{1}{\pi} \theta'_{0, \frac{1}{2}}(\lambda - \lambda_{\frac{1}{2}}) \right) \quad (1.82)$$

The sloution of (1.82) is

$$\rho_s(\lambda) = \rho_0(\lambda) + \frac{1}{N} [\tau(\lambda - \lambda_1) + \tau(\lambda - \lambda_2) + \sigma(\lambda - \lambda_{\frac{1}{2}})] \quad (1.83)$$

where $\tau(\lambda)$ is given by equation (1.78) and $\sigma(\lambda)$ is the solution of

$$\sigma(\lambda) + \frac{1}{2\pi} \int_{-\infty}^{\infty} K(\lambda - \mu) \sigma(\mu) d\mu = -\frac{1}{\pi} \theta'_{0, \frac{1}{2}}(\lambda) \quad (1.84)$$

Given equation (1.58) and deriving $I_{\frac{1}{2}}^{max} = 0$ from equation (1.61), we get

$$2 \arctan \frac{\lambda_{\frac{1}{2}}}{2} = \frac{1}{N} \sum_j \theta_{\frac{1}{2}, 0}(\lambda_{\frac{1}{2}} - \lambda_{0,j}) = \int_{-\infty}^{\infty} \theta_{\frac{1}{2}, 0}(\lambda_{\frac{1}{2}} - \lambda) \rho_s(\lambda) d\lambda \quad (1.85)$$

After some calculations, we get

$$\lambda_{\frac{1}{2}} = \frac{\lambda_1 + \lambda_2}{2} \quad (1.86)$$

1.6 Ground state and low energy excitations

Theorem 1.6.1. *If $\{\lambda_1, \dots, \lambda_N\}$ is a solution of Bethe Ansatz equation (1.41), then $\forall i, j \quad \lambda_i \neq \lambda_j$*

We define

$$Z(\lambda) = \frac{1}{2\pi} [\theta_1(\lambda) - \frac{1}{N} \sum_{l=1}^R \theta_2(\lambda - \lambda_l)] \quad (1.87)$$

$$\sigma(\lambda) = \frac{dZ(\lambda)}{d\lambda} \quad (1.88)$$

Apparently, $Z(\lambda_j) = I_j/N$.

For any consecutive I_j and $I_{j+1} = I_j + 1$, the following relation holds:

$$\frac{Z(\lambda_{j+1}) - Z(\lambda_j)}{\lambda_{j+1} - \lambda_j} = \frac{1}{N\delta\lambda_j} \quad (1.89)$$

In the limit $N \rightarrow \infty$, we obtained

$$\rho(\lambda) = \sigma(\lambda) = a_1(\lambda) - \int_{-\infty}^{\infty} a_2(\lambda - \mu) \rho(\mu) d\mu \quad (1.90)$$

where

$$a_n(\lambda) = \frac{1}{\pi} \frac{n}{\lambda^2 + n^2} \quad (1.91)$$

For ground state

$$\rho_g(\lambda) = a_1(\lambda) - \int_{-\infty}^{\infty} a_2(\lambda - \mu) \rho_g(\mu) d\mu \quad (1.92)$$

To solve (1.92), we use Fourier transformation:

$$\tilde{F}(\omega) = \int_{-\infty}^{\infty} e^{i\omega\lambda} F(\lambda) d\lambda \quad (1.93)$$

$$F(\lambda) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega\lambda} \tilde{F}(\omega) d\omega \quad (1.94)$$

The fourier transformation of $a_n(\lambda)$ is

$$\tilde{a}_n(\omega) = \int_{-\infty}^{\infty} e^{i\omega\lambda} \frac{1}{\pi} \frac{n}{\lambda^2 + n^2} d\lambda = e^{-n|\omega|} \quad (1.95)$$

Take the fourier transformation of (1.92), we obtain

$$\tilde{\rho}_g(\omega) = \frac{1}{2 \cosh \omega} \quad (1.96)$$

Thus the solution of (1.96) is

$$\rho_g(\lambda) = \frac{1}{4 \cosh \frac{\pi\lambda}{2}} \quad (1.97)$$

The density of flipped spins relative to the reference state is

$$\frac{R}{N} = \int_{-\infty}^{\infty} \rho_g(\lambda) d\lambda = \frac{1}{2} \quad (1.98)$$

which means that the magnetization of the ground state is zero. The energy density of the ground state reads

$$e_g = -2\pi \int_{-\infty}^{\infty} a_1(\lambda) \rho_g(\lambda) d\lambda + \frac{1}{2} = \frac{1}{2} - 2 \ln 2 \quad (1.99)$$

1.6.1 Spinon Excitations

We focus on the even N case. The simplest excitation i.e. $R = \frac{N}{2} - 1$. Thus we have

$$\rho^h(\lambda) = \frac{1}{N} \delta(\lambda - \lambda_1^h) + \frac{1}{N} \delta(\lambda - \lambda_2^h) \quad (1.100)$$

Why are there two holes

We suppose $\rho(\lambda) = \rho_g(\lambda) + \delta\rho(\lambda)$. Then (1.90) becomes

$$\delta\rho(\lambda) + \rho^h(\lambda) = - \int_{-\infty}^{\infty} a_2(\lambda - \mu) \delta\rho(\mu) d\mu \quad (1.101)$$

which can be solved by Fourier transformation. We obtain the total spin S_e of this excitation as

$$S_e = -N \int_{-\infty}^{\infty} \delta\rho(\lambda) d\lambda = 1 \quad (1.102)$$

with

$$\delta\rho(\lambda) = \frac{1}{4} \left(\text{sech}\left[\frac{\pi}{2}(\lambda + \lambda_1^h)\right] + \text{sech}\left[\frac{\pi}{2}(\lambda + \lambda_2^h)\right] \right) \quad (1.103)$$

The excitation energy is

$$\Delta E = 2\pi N J \int_{-\infty}^{\infty} a_1(\lambda) \delta\rho(\lambda) d\lambda = \epsilon(\lambda_1^h) + \epsilon(\lambda_2^h) \quad (1.104)$$

where $\epsilon(\lambda)$ is the dressed energy with the definition

$$\epsilon(\lambda) = -2\pi J a_1(\lambda) - \int_{-\infty}^{\infty} a_2(\lambda - \mu) \epsilon(\mu) d\mu \quad (1.105)$$

Solving (1.105) by Fourier transformation, we obtain (similar to (1.92))

$$\epsilon(\lambda) = \frac{-\pi J}{\cosh(\pi\lambda)} = -2\pi J \rho_g(\lambda) \quad (1.106)$$

1.7 Thermodynamics

$$\epsilon_n^0(\lambda) = -2\pi a_n(\lambda) + n\hbar$$

The number of n -string states in the interval $[\lambda, \lambda + d\lambda]$ is

$$N[\rho_n(\lambda) + \rho_n^h(\lambda)]d\lambda \quad (1.107)$$

Then the number of the possible physical states in this interval is

$$d\Omega(\lambda) = \prod_{n=1}^{\infty} \frac{[N(\rho_n(\lambda) + \rho_n^h(\lambda))d\lambda]!}{[N\rho_n(\lambda)d\lambda]! [N\rho_n^h(\lambda)d\lambda]!} \quad (1.108)$$

The entropy in the interval

$$dS(\lambda) = \ln d\Omega(\lambda) \simeq N \sum_{n=1}^{\infty} \{[\rho_n(\lambda) + \rho_n^h(\lambda)] \ln[\rho_n(\lambda) + \rho_n^h(\lambda)] - \rho_n(\lambda) \ln \rho_n(\lambda) - \rho_n^h(\lambda) \ln \rho_n^h(\lambda)\} d\lambda \quad (1.109)$$

Density of free energy

$$f = \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} \epsilon_n^0(\lambda) \rho_n(\lambda) d\lambda - T \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} \{[\rho_n(\lambda) + \rho_n^h(\lambda)] \ln[\rho_n(\lambda) + \rho_n^h(\lambda)] - \rho_n(\lambda) \ln \rho_n(\lambda) - \rho_n^h(\lambda) \ln \rho_n^h(\lambda)\} d\lambda \quad (1.110)$$

To get a thermal equilibrium state

$$\frac{\delta f}{\delta \rho_n(\lambda)} = 0 \quad (1.111)$$

which leads to

$$\sum_{n=1}^{\infty} \int_{-\infty}^{\infty} \{\epsilon_n^0(\lambda) \delta \rho_n(\lambda) - T \ln[1 + \eta_n(\lambda)] \delta \rho_n(\lambda) - T \ln[1 + \eta_n^{-1}(\lambda)] \delta \rho_n^h(\lambda)\} d\lambda = 0 \quad (1.112)$$

where

$$\eta_n(\lambda) = \frac{\rho_n^h(\lambda)}{\rho_n(\lambda)} \quad (1.113)$$

Note that $\rho_n(\lambda)$ and $\rho_n^h(\lambda)$ are related by

$$\delta \rho_n^h(\lambda) = - \sum_{m=1}^{\infty} \int_{-\infty}^{\infty} A_{m,n}(\lambda - \mu) \delta \rho_m(\mu) d\mu \quad (1.114)$$

Substituting (1.114) to (1.112), we get

$$\ln[1 + \eta_n(\lambda)] = \frac{\epsilon_n^0(\lambda)}{T} + \sum_{m=1}^{\infty} \int_{-\infty}^{\infty} A_{m,n}(\lambda - \mu) \ln[1 + \eta_m^{-1}(\mu)] d\mu \quad (1.115)$$

We define the integral operator $[n]$ as

$$[n]F(\lambda) = \int_{-\infty}^{\infty} a_n(\lambda - \mu) F(\mu) d\mu \quad (1.116)$$

Note that under fourier transformation $[n]$ becomes a multiplier $\exp(-n|\omega|/2)$.
We have

$$[m][n] = [m+n] \quad (1.117)$$

It can be easily verified in ω space. Further, we define

$$\begin{aligned} \hat{\mathbf{A}}_{m,n} &= [m+n] + 2[m+n-2] + \cdots + 2[|m-n|+2] + [|m-n|] \\ \hat{\mathbf{G}} &= \frac{[1]}{[0] + [2]} \end{aligned} \quad (1.118)$$

Where the kernel of the operator $\hat{\mathbf{G}}$ is

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\lambda\omega} \frac{e^{-\frac{1}{2}|\omega|}}{1 + e^{-|\omega|}} d\omega = \frac{1}{2\cosh(\pi\lambda)} = \rho_g(\lambda) \quad (1.119)$$

The operator identities

$$\hat{\mathbf{G}}[\hat{\mathbf{A}}_{m,n+1} + \hat{\mathbf{A}}_{m,n-1}] = -\delta_{m,n} + \hat{\mathbf{A}}_{m,n}, \quad n > 1 \quad (1.120)$$

$$\hat{\mathbf{G}}\hat{\mathbf{A}}_{m,2} = -\delta_{1,m} + \hat{\mathbf{A}}_{1,m} \quad (1.121)$$

We can then rewrite (1.115) as

$$\ln(1 + \eta_n(\lambda)) = \frac{\epsilon_n^0(\lambda)}{T} + \sum_{m=1}^{\infty} \hat{\mathbf{A}}_{n,m} \ln(1 + \eta_m^{-1}(\lambda)) \quad (1.122)$$

$\hat{\mathbf{G}} \rightarrow$ (1.122) with $n = n-1$ and $n = n+1$

$$\begin{aligned} &\hat{\mathbf{G}}[\ln(1 + \eta_{n+1}(\lambda)) + \ln(1 + \eta_{n-1}(\lambda))] \\ &= \frac{\epsilon_n^0(\lambda)}{T} - \ln(1 + \eta_n^{-1}(\lambda)) + \sum_{m=1}^{\infty} \hat{\mathbf{A}}_{n,m} \ln(1 + \eta_m^{-1}(\lambda)) \end{aligned} \quad (1.123)$$

Combining (1.122) and (1.123), we get

$$\ln \eta_n(\lambda) = \hat{\mathbf{G}}[\ln(1 + \eta_{n+1}(\lambda)) + \ln(1 + \eta_{n-1}(\lambda))] \quad (1.124)$$

Applying $\hat{\mathbf{G}}$ with $n = 2$, we obtain

$$\ln \eta_1(\lambda) = -\frac{2\pi\rho_g(\lambda)}{T} + \hat{\mathbf{G}} \ln(1 + \eta_2(\lambda)) \quad (1.125)$$

We obtain

$$f = -T \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} a_n(\lambda) \ln[1 + \eta_n^{-1}(\lambda)] d\lambda \quad (1.126)$$

From (1.122), we know that

$$\hat{\mathbf{G}} \ln(1 + \eta_1(\lambda)) = \frac{1}{T} \hat{\mathbf{G}} \epsilon_1^0(\lambda) + \sum_{m=1}^{\infty} [m] \ln(1 + \eta_m^{-1}(\lambda)) \quad (1.127)$$

Let $\lambda = 0$, we get

$$\begin{aligned} \sum_{m=1}^{\infty} \int_{-\infty}^{\infty} a_m(\lambda) \ln[1 + \eta_m^{-1}(\lambda)] d\lambda \\ = \frac{2 \ln 2 - \frac{1}{2h}}{T} + \int_{-\infty}^{\infty} \rho_g(\lambda) \ln[1 + \eta_1(\lambda)] d\lambda \end{aligned} \quad (1.128)$$

Substituting (1.128) into (1.126), we finally get the expression for the energy

$$\frac{F}{N} = e_g - T \int_{-\infty}^{\infty} \rho_g(\lambda) \ln[1 + \eta_1(\lambda)] d\lambda \quad (1.129)$$

Below, let us consider the low energy limit $T \rightarrow 0$ and $h \rightarrow h$. Then $\eta_1 = 0$ and η_n become constants. In this case, (1.124) (1.125) are reduced to

$$\eta_n^2 = (1 + \eta_{n+1})(1 + \eta_{n-1}), \quad n > 1 \quad (1.130)$$

The general solution of the above equations is

$$\eta_n = \left(\frac{bz^n - b^{-1}z^{-n}}{z - z^{-1}} \right)^2 - 1 \quad (1.131)$$

where b and z are determined by $\eta_1 = 0$ and limitation condition as $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \frac{\ln \eta_n}{n} = \frac{h}{T} \quad (1.132)$$

In the case of $h = 0$, we have $\eta_n(h = 0) = n^2 - 1$. Let $\eta_n = \exp[-\epsilon_n/T]$, then $\epsilon_1 \sim T^0$. Comparing the T^{-1} terms of (1.122) we have

$$\epsilon_1(\lambda) = -\epsilon_1^0(\lambda) - \int_{-\infty}^{\infty} a_2(\lambda - \mu) \epsilon_1(\mu) d\mu \quad (1.133)$$

Equivalently

$$\epsilon_1(\lambda) = -2\pi J a_1(\lambda) + H - \int_{-\infty}^{\infty} a_2(\lambda - \mu) \epsilon_1(\mu) d\mu \quad (1.134)$$

where $a_n = \frac{1}{2\pi} \frac{n}{\lambda^2 + \frac{n^2}{4}}$. (This definition is different from the precedent content of note.)

$$\epsilon_1(\lambda) = -2\pi J a_1(\lambda) + H - \int_{-Q}^Q a_2(\lambda - \mu) \epsilon_1(\mu) d\mu \quad (1.135)$$

$$f_0 = \int_{-Q}^Q a_1(\mu) \epsilon_1(\mu) d\mu \quad (1.136)$$

Where $\epsilon_1(\pm Q) = 0$, by let $\epsilon_1(0) = 0$ we get $H_s = 4J$.

At zero temperature, we consider $H \rightarrow H_s$. In that case, Q is small. We expand $\epsilon_1(\lambda)$ in terms of small λ .

$$\epsilon_1(\lambda) = -2\pi J a_1(\lambda) + H - \frac{1}{\pi} \int_{-Q}^Q \epsilon_1(\lambda) d\lambda \approx -2\pi J a_1(\lambda) + H - \frac{2Q(H - H_s)}{\pi} \quad (1.137)$$

Then how could we get $Q = \sqrt{\frac{H_s - H}{16J}}$. Using $\epsilon_1(Q) = 0$?

The zero temperature dressed energy

$$f_0 \approx \int_{-Q}^Q a_1(\mu) \epsilon_1(\mu) d\mu \approx \frac{4J}{\pi} \frac{(1 - 4Q^2) \arctan 2Q - 2Q}{1 + 4Q^2} + O(Q^4) \quad (1.138)$$

We can derive some emergent quantities from free energy f

$$M^z = \frac{1}{2} - \frac{\partial f}{\partial H} \quad \chi = \frac{\partial^2 f}{\partial H^2} \quad c_V = -T \frac{\partial^2 f}{\partial T^2} \quad (1.139)$$

Substituting (1.138) to (1.139), we get

$$M^z = \frac{1}{2} - \frac{\partial f_0}{\partial H} = \frac{1}{2} - \frac{2}{\pi} \left(1 - \frac{H}{H_s}\right)^{1/2} \quad (1.140)$$

$$\chi = \frac{\partial M^z}{\partial H} = \frac{1}{2\pi} \left(\frac{1}{J(H_s - H)} \right)^{1/2} \quad (1.141)$$

We give the scaling form of (1.140)

$$1 - \frac{M^z}{M_s} = D \left(1 - \frac{H}{H_s}\right)^{1/\delta} = \frac{4}{\pi} \left(1 - \frac{H}{H_s}\right)^{1/2} \quad (1.142)$$

1.7.1 Luttinger Liquid

The thermodynamic Bethe ansatz (TBA) equations

$$\epsilon_n^+ = \epsilon_n^0 - \sum_m A_{m,n} * \epsilon_n^- \quad (\text{TBA, 142.5})$$

with $n = 1, 2, \dots, \infty$. The $*$ denotes convolution and $\epsilon^\pm = \pm T \ln(1 + e^{\pm \epsilon_n/T})$. The deriving term is $\epsilon_n^0 = -2\pi J a_n(\lambda) + nH = -\frac{nJ}{\lambda^2 + n^2/4} nH$.

Luttinger liquid is a collective motion of particle-hole excitations near two Fermi points at low temperature. The condition $|H - H_s|/T \gg 1$ is necessary for this excitation. We rewrite the low temperature TBA equation $\epsilon_1^T(\lambda) = \epsilon_1(\lambda) + \eta(\lambda)$, where $\epsilon_1(\lambda)$ is the zero temperature dressed energy (1.134), η can be regarded as a leading correction to the temperature.

We give first

$$\epsilon_1^T(\lambda) = -2\pi J a_1(\lambda) + H + T \int_{-\infty}^{\infty} a_2(\lambda - \mu) \ln \left(1 + e^{\frac{-|\epsilon_1^T(\mu)|}{T}}\right) d\mu - \int_{-Q}^Q a_2(\lambda - \mu) \epsilon_1^T(\mu) d\mu \quad (1.143)$$

And then

$$\begin{aligned} \epsilon_1^T(\lambda) &= \epsilon_1(\lambda) + \eta(\lambda) \\ &= -2\pi J a_1(\lambda) + H - \int_{-Q}^Q a_2(\lambda - \mu) \epsilon_1(\mu) d\mu + \eta(\lambda) \\ &= -2\pi J a_1(\lambda) + H - \int_{-Q}^Q a_2(\lambda - \mu) (\epsilon_1^T - \eta(\mu)) d\mu + \eta(\lambda) \end{aligned} \quad (1.144)$$

Thus we have

$$\eta(\lambda) = T \int_{-\infty}^{\infty} a_2(\lambda - \mu) \ln \left(1 + e^{\frac{-|\epsilon_1^T(\mu)|}{T}} \right) d\mu - \int_{-Q}^Q a_2(\lambda - \mu) \eta(\mu) d\mu \quad (1.145)$$

As $T \rightarrow 0$, only the zeros of ϵ_1^T contribute to the first integration. We expanding ϵ_1^T at $\lambda = Q$

$$\epsilon_1^T(\lambda) = t(\lambda - Q) + O((\lambda - Q)^2) \quad (1.146)$$

where $t = \frac{d\epsilon_1^T(\lambda)}{d\lambda}|_{\lambda=Q}$. Then the integration becomes

$$I = \frac{\pi^2 T^2}{6t} [a_2(\lambda + Q) + a_2(\lambda - Q)] \quad (1.147)$$

In that case

$$\eta(\lambda) = \frac{\pi^2 T^2}{6t} [a_2(\lambda + Q) + a_2(\lambda - Q)] - \int_{-Q}^Q a_2(\lambda - \mu) \eta(\mu) d\mu \quad (1.148)$$

At zero temperature, the free energy $f(T, H)$ is given by

$$f_0(0, H) = \int_{-Q}^Q a_1(\lambda) \epsilon_1(\lambda) d\lambda \quad (1.149)$$

Under such conditions, the free energy is given by

$$f(T, H) = -T \int_{-\infty}^{\infty} a_1(\lambda) \ln \left(1 + e^{\frac{-\epsilon_1^T(\lambda)}{T}} \right) d\lambda \quad (1.150)$$

It follows that

$$\begin{aligned} f - f_0 &= -T \int_{-\infty}^{\infty} a_1(\lambda) \ln \left(1 + e^{\frac{-\epsilon_1^T(\lambda)}{T}} \right) d\lambda - \int_{-Q}^Q a_1(\lambda) \epsilon_1(\lambda) d\lambda \\ &= -T \int_{-\infty}^{\infty} a_1(\lambda) \ln \left(1 + e^{\frac{-|\epsilon_1^T(\lambda)|}{T}} \right) d\lambda + \int_{-Q}^Q a_1(\lambda) \eta(\lambda) d\lambda \\ &= -\frac{\pi^2 T^2}{3t} a_1(Q) + \int_{-Q}^Q a_1(\lambda) \eta(\lambda) d\lambda \end{aligned} \quad (1.151)$$

We can express the free energy as

$$f = f_0 - \frac{\pi^2 T^2}{3t} a_1(Q) + \int_{-Q}^Q a_1(\lambda) \eta(\lambda) d\lambda \quad (1.152)$$

To calculate the last term of (1.152), we first give

$$\rho_0(\lambda) = a_1(\lambda) - \int_{-Q}^Q a_2(\lambda - \mu) \rho_0(\mu) d\mu \quad (1.153)$$

Note that (1.148) has the same structure as (1.153) and we have the theorem such that:
If we have

$$f(k) = f_0(k) - \int_{-Q}^Q a_2(k - \lambda) f(\lambda) d\lambda \quad (1.154)$$

$$g(k) = g_0(k) - \int_{-Q}^Q a_2(k - \lambda)g(\lambda)d\lambda \quad (1.155)$$

then

$$\int_{-Q}^Q f(\lambda)g_0(\lambda)d\lambda = \int_{-Q}^Q g(\lambda)f_0(\lambda)d\lambda \quad (1.156)$$

Using this theorem we can obtain

$$\int_{-Q}^Q \frac{\pi^2 T^2}{6t} [(a_2(\lambda + Q) + a_2(\lambda - Q))] \rho_0(\lambda) d\lambda = \int_{-Q}^Q a_1(\lambda) \eta(\lambda) d\lambda \quad (1.157)$$

Assign $\lambda = \pm Q$ to (1.153), we can substitute LHS of (1.157) with

$$\int_{-Q}^Q [(a_2(\lambda + Q) + a_2(\lambda - Q))] \rho_0(\lambda) d\lambda = 2a_1(Q) - 2\rho_0(Q) \quad (1.158)$$

We then obtain

$$\int_{-Q}^Q a_1(\lambda) \eta(\lambda) d\lambda = \frac{\pi^2 T^2}{6t} [2a_1(Q) - 2\rho_0(Q)] \quad (1.159)$$

Finally, we simplify (1.152) to the following form

$$f = f_0 - \frac{\pi^2 T^2}{3t} \rho_0(Q) \quad (1.160)$$

We then consider the specific heat at TTL region.

$$c_v = -T \frac{\partial^2 f}{\partial T^2} = \frac{\pi T}{3v_s} \propto T^\alpha \quad (1.161)$$

where v_s is defined as sound velocity

$$v_s = \frac{1}{2\pi} \frac{d\epsilon_1(\lambda)/d\lambda}{\rho_0(\lambda)} \Big|_{\lambda=Q} = \frac{1}{2\pi} \frac{t}{\rho_0(Q)} \quad (1.162)$$

We start from the Hamiltonian of Luttinger liquid

$$H = \frac{\hbar}{2\pi} \int dx \left(\frac{v_s K_s}{\hbar^2} (\pi \Pi(x))^2 + \frac{v_s}{K_s} (\delta \phi(x))^2 \right) \quad (1.163)$$

1.8 Quantum criticality

The free energy and TBA equations are as Below

$$f = -T \int a_1(\lambda) \ln \left(1 + e^{\frac{-\epsilon_1(\lambda)}{T}} \right) d\lambda \quad (1.164)$$

$$\epsilon_1(\lambda) = -2\pi J a_1(\lambda) + H + T \int a_2(\lambda - \mu) \ln \left(1 + e^{\frac{-\epsilon_1(\mu)}{T}} \right) d\mu \quad (1.165)$$

We take the expansion of the kernel function

$$a_n(\lambda) = \frac{1}{2\pi} \frac{n}{\lambda^2 + n^2/4} \approx \frac{2}{n\pi} \left(1 - \frac{4}{n^2} \lambda^2 + \dots \right) \quad (1.166)$$

We denote

$$b_1 = T \int \ln \left(1 + e^{-\frac{\epsilon_1(\mu)}{T}} \right) d\mu \quad (1.167)$$

$$b_2 = T \int \mu^2 \ln \left(1 + e^{-\frac{\epsilon_1(\mu)}{T}} \right) d\mu \quad (1.168)$$

Then we can rewrite (1.164) and (1.165) as

$$f \approx -\frac{2}{\pi} b_1 + \frac{8}{\pi} b_2 \quad (1.169)$$

$$\epsilon_1(\lambda) \approx \left(16J - \frac{b_1}{\pi} \right) \lambda^2 - 4J + H + \frac{b_1}{\pi} - \frac{b_2}{\pi} \quad (1.170)$$

Substituting (1.170) to (1.167) (1.168) and integrating by parts, we obtain the following close forms after recognizing the integral form of polylogarithm function.

$$b_1 = -\frac{\sqrt{\pi} T^{\frac{3}{2}}}{(16J - \frac{b_1}{\pi})^{\frac{1}{2}}} f_{\frac{3}{2}}^{A_0}, \quad (1.171)$$

$$b_2 = -\frac{1}{2} \frac{\sqrt{\pi} T^{\frac{5}{2}}}{(16J - \frac{b_1}{\pi})^{\frac{3}{2}}} f_{\frac{5}{2}}^{A_0} \quad (1.172)$$

Where $A_0 = 4J - H - \frac{b_1}{\pi} + \frac{b_2}{\pi}$ and $f_n^{A_0} = Li_n(-e^{\frac{A_0}{T}})$. Using this expressions, we rewrite dressed energy and free energy (1.170) (1.169) as

$$\begin{aligned} \epsilon_1(\lambda) = & \left(16J - \frac{b_1}{\pi} \right) \lambda^2 - 4J + H - \frac{1}{4\sqrt{\pi J}} \frac{T^{\frac{3}{2}}}{(1 - \frac{b_1}{16\pi J})^{\frac{1}{2}}} f_{\frac{3}{2}}^{A_0} \\ & + \frac{1}{8\sqrt{\pi J}(16J)} \frac{T^{\frac{5}{2}}}{(1 - \frac{b_1}{16\pi J})^{\frac{3}{2}}} f_{\frac{5}{2}}^{A_0} \end{aligned} \quad (1.173)$$

$$f = \frac{T^{\frac{3}{2}}}{2\sqrt{\pi J}(1 - \frac{b_1}{16\pi J})^{\frac{1}{2}}} f_{\frac{3}{2}}^{A_0} - \frac{T^{\frac{5}{2}}}{16J\sqrt{\pi J}(1 - \frac{b_1}{16\pi J})^{\frac{3}{2}}} f_{\frac{5}{2}}^{A_0} \quad (1.174)$$

The b_1 and b_2 above in f , ϵ_1 and A_0 should be understood as

$$b_1^s = -\frac{\sqrt{\pi} T^{\frac{3}{2}}}{4\sqrt{J}} Li_{\frac{3}{2}} \left(-e^{\frac{H_s - H}{T}} \right)$$

$$b_2^s = -\frac{1}{2} \frac{\sqrt{\pi} T^{\frac{5}{2}}}{(16J)^{\frac{3}{2}}} Li_{\frac{5}{2}} \left(-e^{\frac{H_s - H}{T}} \right)$$

The magnetization is given by

$$M^z = \frac{1}{D_m} \frac{-T^{\frac{1}{2}}}{2\sqrt{\pi J}} f_{\frac{1}{2}}^s \left(1 - \frac{T}{8J} f_{\frac{3}{2}}^s / f_{\frac{1}{2}}^s\right) + O((T/J)^2) \quad (1.175)$$

$$D_m = 1 - \frac{T^{\frac{1}{2}}}{\sqrt{16\pi J}} f_{\frac{1}{2}}^s + \frac{T^{\frac{3}{2}}}{2\sqrt{\pi}(16J)^{\frac{3}{2}}} f_{\frac{3}{2}}^s \quad (1.176)$$

where $f_n^s = Li_n(-e^{\frac{4J-H}{T}})$.

1.8.1 Scaling function

In the vicinity of the critical point H_s and $|H - H_s|/T \ll 1$, we can obtain the scaling forms directly from the free energy (1.174) with an extra condition $J/T \gg 1$.

$$f \approx \frac{T^{\frac{3}{2}}}{2\sqrt{\pi J}} Li_{\frac{3}{2}} \left(-e^{\frac{H_s-H}{T}}\right) \quad (1.177)$$

It follows that the scaling forms of the Magnetization and susceptibility

$$M^z = \frac{1}{2} + \frac{T^{\frac{1}{2}}}{2\sqrt{\pi J}} Li_{\frac{1}{2}} \left(-e^{\frac{4J-H}{T}}\right) = \frac{1}{2} + T^{\frac{1}{2}} M(\Delta H/T) \quad (1.178)$$

$$\chi = \frac{\partial M^z}{\partial H} = -\frac{1}{2\sqrt{\pi J T}} Li_{-\frac{1}{2}} \left(-e^{\frac{4J-H}{T}}\right) = T^{-\frac{1}{2}} G(\Delta H/T) \quad (1.179)$$

In the above equations the functions $M(x) = \frac{1}{2\sqrt{\pi J}} f_{\frac{1}{2}}^s(x)$, $G(x) = -\frac{1}{2\sqrt{\pi J}} f_{-\frac{1}{2}}^s(x)$ are dimensionless scaling functions. Here we denote $f_n^s(\Delta/T) = Li_n(-e^{\frac{\Delta}{T}})$ and $\Delta = H_s - H = 4J - H$. Similarly, the scaling function of the specific heat is given by

$$\begin{aligned} c_v &= \frac{\partial s}{\partial T} = -T \frac{\partial^2 f}{\partial T^2} \\ &= \sqrt{\frac{T}{\pi J}} \left[-\frac{3}{8} Li_{\frac{3}{2}} \left(-e^{\frac{\Delta H}{T}}\right) + \frac{1}{2} \left(\frac{\Delta H}{T}\right) Li_{\frac{1}{2}} \left(-e^{\frac{\Delta H}{T}}\right) - \frac{1}{2} \left(\frac{\Delta H}{T}\right)^2 Li_{-\frac{1}{2}} \left(-e^{\frac{\Delta H}{T}}\right) \right] \\ &= T^{\frac{1}{2}} C(\Delta H/T) \end{aligned} \quad (1.180)$$

Furthermore, we can also get the scaling form of the Wilson Ratio in critical region

$$R_W = \frac{4}{3} \left(\frac{\pi k_B}{g\mu_B}\right)^2 \frac{f_{-\frac{1}{2}}^s}{\frac{3}{4} f_{\frac{3}{2}}^s - \frac{\Delta H}{T} f_{\frac{1}{2}}^s + \left(\frac{\Delta H}{T}\right)^2 f_{-\frac{1}{2}}^s} \approx \left(\frac{4\pi k_B}{3g\mu_B}\right)^2 \frac{f_{-\frac{1}{2}}^s}{f_{\frac{3}{2}}^s} \quad (1.181)$$

1.8.2 Extrem value

Substituting $y = \Delta H/T$ to (1.180), we get

$$C_v = \sqrt{\frac{T}{\pi J}} \left[-\frac{3}{8} Li_{\frac{3}{2}}(-e^y) + \frac{1}{2} y Li_{\frac{1}{2}}(-e^y) - \frac{1}{2} y^2 Li_{-\frac{1}{2}}(-e^y) \right] \quad (1.182)$$

To get the extrem values of C_v , we do differentiation over H

$$\frac{\partial C_v}{\partial H} = \frac{\partial C_v}{\partial y} \frac{\partial y}{\partial H} = -\frac{1}{T} \left[\frac{\partial C_v}{\partial y} \right] \quad (1.183)$$

where

$$\frac{\partial C_v}{\partial y} = \sqrt{\frac{T}{\pi J}} \left[\frac{1}{8} Li_{\frac{1}{2}}(-e^y) - \frac{1}{2} y Li_{-\frac{1}{2}}(-e^y) - \frac{1}{2} y^2 Li_{-\frac{3}{2}}(-e^y) \right] \quad (1.184)$$

In that case, the extrem value equation is

$$\frac{1}{8} Li_{\frac{1}{2}}(-e^y) - \frac{y}{2} Li_{-1/2}(-e^y) - \frac{1}{2} y^2 Li_{-\frac{3}{2}}(-e^y) = 0 \quad (1.185)$$

Similarly, the extrem value equation of M^z over T reads

$$\begin{aligned} \frac{\partial M^z}{\partial T} &= \frac{1}{2\sqrt{\pi J T}} \left[\frac{1}{2} Li_{\frac{1}{2}}(-e^y) - y Li_{-\frac{1}{2}}(-e^y) \right] \\ &= 0 \end{aligned} \quad (1.186)$$

Equivalently

$$\frac{1}{2} Li_{\frac{1}{2}}(-e^y) - y Li_{-\frac{1}{2}}(-e^y) = 0 \quad (1.187)$$

1.9 Entanglement and Quantum Fisher information of spinons

- Pure state and Mixed state.

We choose $\{\phi_i(x)\}$ as our basis.

- Pure state: $\psi(x) = c_1 \phi_1(x) + c_2 \phi_2(x)$.

We measure to get $\phi_1(x)$ with probability $|c_1|^2$ and $\phi_2(x)$ with probability $|c_2|^2$.

- Mixed states. $\{\psi_i\} \rightarrow \{p_i\}$ with $\sum_i p_i = 1$
 $\hat{\rho} = \sum_i p_i |\psi_i\rangle \langle \psi_i|$

- Quantum entanglement Witness.

1. A pure quantum state $|\psi\rangle \in H_1 \otimes \cdots \otimes H_N$ is called a **product state** or **seprable** if we can find states $|\phi_j\rangle \in H_j$ for $j = 1, \dots, N$ such that

$$|\psi\rangle = |\phi_1\rangle \otimes \cdots \otimes |\phi_N\rangle$$

2. A mixed state ρ is called a **(full) separable** state if it can be written as a covex combination of some product states $\rho_j^1 \otimes \cdots \rho_j^N$.

$$\rho = \sum_j p_j \rho_j^1 \otimes \cdots \otimes \rho_j^N,$$

where $p_j \geq 0$ and $\sum_j p_j = 1$. If a state is not separable then it is **entangled**.

- Multiparticle Entanglement

1. A pure state $|\psi\rangle \in H_1 \otimes \cdots \otimes H_N$ is **k-producible** if it can be written as $|\psi\rangle = \otimes_{j=1}^M |\phi_j\rangle$, where $|\phi_j\rangle$ is a state of $N_j \leq k$ particles (such that $\sum_{j=1}^M N_j = N$)
2. Similarly, a mixed state is **k-producible**, if it can be written as convex mixture of l-producible pure states, i.e. $\rho = \sum_l p_l |\psi_{k_l-prod}\rangle \langle \psi_{k_l-prod}|$, where $k_l \leq k, p_l > 0, \sum_l p_l = 1$.
3. If a quantum state is k-producible, not k-1-producible, then it is k-particle entangled, the Entanglement depth is larger than $k-1$.

1.9.1 Quantum Fisher Information

We define the **symmetry logarithmic derivation (SLD) operator** as

$$\frac{\partial \rho(\theta)}{\partial \theta} = \frac{1}{2} \{ \rho(\theta), L \} = \frac{1}{2} [\rho(\theta) L + L \rho(\theta)]. \quad (1.188)$$

In that case, the quantum fisher information (QFI) for a one-parameter quantum state $\rho(\theta)$ is defined as

$$\mathcal{F}_Q[\rho(\theta)] = \text{Tr}(\rho(\theta) L^2) \quad (1.189)$$

Suppose that $\rho(\theta) = \sum_k \lambda_k(\theta) |k(\theta)\rangle \langle k(\theta)|$ (spectrum decomposition) then

$$L = \sum_k' \frac{\partial_\theta \lambda_k}{\lambda_k} |k\rangle \langle k| + 2 \sum_{k,k'}' \frac{\lambda_k - \lambda_{k'}}{\lambda_k + \lambda_{k'}} \langle \partial_\theta k | k' \rangle |k\rangle \langle k'| \quad (1.190)$$

where $\sum'$ indicates that the summation requires $\lambda_k \neq 0$ and $\lambda_k + \lambda_{k'} \neq 0$. Substituting (1.190) to (1.189) we get

$$\mathcal{F}_Q[\rho(\theta)] = \sum_k' \frac{(\partial_\theta \lambda_k(\theta))^2}{\lambda_k(\theta)} + 2 \sum_{k,k'}' \frac{(\lambda_k(\theta) - \lambda_{k'}(\theta))^2}{\lambda_k(\theta) + \lambda_{k'}(\theta)} |\langle \partial_\theta k(\theta) | k'(\theta) \rangle|^2 \quad (1.191)$$

In particular, if a pure state $\rho(\theta) = |\psi(\theta)\rangle \langle \psi(\theta)|$ then

$$\mathcal{F}_Q[|\psi(\theta)\rangle] = 4 |\langle \partial_\theta \psi(\theta) | \partial_\theta \psi(\theta) \rangle - 4 |\langle \partial_\theta \psi(\theta) | \psi(\theta) \rangle|^2 \quad (1.192)$$

To prove (1.192) we should use $\sum_\perp |\psi^\perp\rangle \langle \psi^\perp| + |\psi\rangle \langle \psi| = 1$. (see my script note)
We give the Quantum Fisher Information inequality.

$$F_Q[\rho_{sep}] \leq \sum_k p_k F_Q[|\psi_{sep}^{(k)}\rangle] = \sum_{i,k} p_k F_Q[|\psi_i^{(k)}\rangle] \leq N(h_{max} - h_{min})^2 \quad (1.193)$$

Where $F_Q[|\psi_i^{(k)}\rangle] \leq 4(\Delta h_i)^2$, $4(\Delta h_i)^2 \leq (h_{max} - h_{min})^2$

1.9.2 Kubo formula and QFI

The Hamiltonian of the system is

$$H(t) = H_0 + V(t) \quad (1.194)$$

where $V(t) = f(t)\theta(t - t_0)\hat{\mathbf{A}}$.

We change to the interaction picture

$$\hat{\mathbf{O}}_I(t) = e^{iH_0t}\hat{\mathbf{O}}_se^{-iH_0t} \quad (1.195)$$

$$|\psi(t)\rangle_I = e^{iH_0t}|\psi(t)\rangle_s \quad (1.196)$$

After solving the equation of motion, we get

$$|\psi(t)\rangle_I = U_I(t, t_0)|\psi(t_0)\rangle_I \quad (1.197)$$

where $U_I(t, t_0) \approx 1 - i \int_{t_0}^t dt' V_I(t')$ After some calculations, we get

$$\delta \langle B \rangle(t) = \int_{-\infty}^{\infty} dt' \chi_{BA}(t - t') f(t') \quad (1.198)$$

where

$$\chi_{BA}(t - t') = -i\theta(t - t') \langle [B_I(t), A_I(t')] \rangle_0 \quad (1.199)$$

By applying Fourier transformation we get

$$\begin{aligned} \delta \langle B \rangle(\omega) &= \int dt e^{i\omega t} \delta \langle B \rangle(t) \\ &= \int dt e^{i\omega t} \int dt' \chi_{BA}(t - t') f(t') \\ &= \int dt e^{i\omega(t-t')} \chi_{BA}(t - t') \int dt' e^{i\omega t'} f(t') \\ &= \chi_{BA}(\omega) f(\omega) \end{aligned} \quad (1.200)$$

And

$$\chi_{BA}(\omega) = -i \int_0^{\infty} dt e^{i\omega t} \langle [B_I(t), A_I(0)] \rangle_0 \quad (1.201)$$

We take conjugate of $\chi_{BA}(\omega)$ and get

$$\chi_{BA}^* = \chi_{B^\dagger A^\dagger}(-\omega) \quad (1.202)$$

We can see the real part $Re(\chi_{BA}) = \frac{1}{2}(\chi_{BA} + \chi_{BA}^*) = \frac{1}{2}(\chi_{BA}(\omega) + \chi_{BA}(-\omega))$ is an even function of ω and the imaginary part an odd function. We write

$$\chi_{BA}(\omega) = \chi'_{BA}(\omega) + i\chi''_{BA}(\omega) \quad (1.203)$$

with $\chi'_{BA}(\omega)$ and $\chi''_{BA}(\omega)$ the real part and imaginary part.

We use Lehmann Representation two-point correlation function in equilibrium

$$\begin{aligned}
\langle [A_I(t), A] \rangle_0 &= \sum_{\lambda, \lambda'} p_\lambda \langle \lambda | e^{iH_0 t} A e^{-iH_0 t} | \lambda' \rangle \langle \lambda' | A | \lambda \rangle - \sum_{\lambda, \lambda'} p_{\lambda'} \langle \lambda' | A | \lambda \rangle \langle \lambda | e^{iH_0 t} A e^{-iH_0 t} | \lambda' \rangle \\
&= \sum_{\lambda, \lambda'} (p_\lambda - p_{\lambda'}) |A_{\lambda, \lambda'}|^2 e^{i(E_\lambda - E_{\lambda'})t} e^{-\eta t}
\end{aligned} \tag{1.204}$$

Thus we get the fourier transformation of Lehman Representation:

$$\begin{aligned}
\chi_{AA}(\omega) &= -i \int_0^\infty dt e^{i\omega t} \langle [A_I(t), A_I(0)] \rangle_0 \\
&= \sum_{\lambda, \lambda'} (p_\lambda - p_{\lambda'}) |A_{\lambda, \lambda'}|^2 (-i) \int_0^\infty dt e^{i\omega t} e^{i(E_\lambda - E_{\lambda'})t} e^{-\eta t} \quad (\eta \rightarrow 0^+) \\
&= - \sum_{\lambda, \lambda'} (p_\lambda - p_{\lambda'}) |A_{\lambda, \lambda'}|^2 [P(\frac{1}{\omega + E_\lambda - E_{\lambda'}}) - i\pi\delta(\omega + E_\lambda - E_{\lambda'})]
\end{aligned} \tag{1.205}$$

Fluctuaion-Dissipation theorem (Imaginary part - DSF relation)

We write $\chi_{AA}(\omega)$ As

$$\chi_{AA}(\omega) = \chi'_{AA}(\omega) + i\chi''_{AA}(\omega) \tag{1.206}$$

We give the imaginary part $\chi_{AA}(\omega)$ as :

$$\chi''_{AA}(\omega) = \frac{1}{2} \int_{-\infty}^\infty dt e^{i\omega t} \langle AA_I(t) - A_I(t)A \rangle_0 \tag{1.207}$$

We define

$$S_A(\omega) = \int_{-\infty}^\infty dt e^{i\omega t} \langle A_I(t)A \rangle_0 \tag{1.208}$$

and note that

$$\langle AA_I(t) \rangle_0 = \langle A_I(t - i\beta)A \rangle_0 \tag{1.209}$$

from which, we get

$$S_A(-\omega) = e^{-\beta\omega} S_A(\omega) \tag{1.210}$$

We define the **Dynamical Structure Factor** $S_{AA}(\omega)$

$$S_{AA}(\omega) = \frac{S_A(\omega) + S_A(-\omega)}{2} = \frac{1}{2}(1 + e^{-\beta\omega})S_A(\omega) \tag{1.211}$$

Consequently, we rewrite (1.207) to

$$\begin{aligned}
\chi''_{AA}(\omega) &= \frac{1}{2}(e^{-\beta\omega} - 1)S_A(\omega) \\
&= -\tanh\left(\frac{\beta\omega}{2}\right)S_{AA}(\omega)
\end{aligned} \tag{1.212}$$

The Fisher Information

$$\mathcal{F}_Q[\rho_0 : A] = -\frac{2}{\pi} \int_{-\infty}^{\infty} d\omega \tanh\left(\frac{\beta\omega}{2}\right) \chi''_{AA}(\omega) \quad (1.213)$$

$$\mathcal{F}_Q[\rho_0 : A] = \frac{2}{\pi} \int_{-\infty}^{\infty} d\omega \tanh^2\left(\frac{\beta\omega}{2}\right) S_{AA}(\omega) \quad (1.214)$$

Using Lehmann representation (1.205), we have

$$\chi''_{AA}(\omega) = -\pi \sum_{\lambda, \lambda'} (p_{\lambda} - p_{\lambda'}) |A_{\lambda, \lambda'}|^2 \delta(\omega + E_{\lambda} - E_{\lambda'}) \quad (1.215)$$

substituting which Fisher Information (1.213) becomes

$$\begin{aligned} \mathcal{F}_Q[\rho_0 : A] &= -\frac{2}{\pi} \int_{-\infty}^{\infty} d\omega \tanh\left(\frac{\beta\omega}{2}\right) \chi''_{AA}(\omega) \\ &= 2 \sum_{\lambda, \lambda'} (p_{\lambda} - p_{\lambda'}) |A_{\lambda, \lambda'}|^2 \tanh\left(\frac{\beta(E_{\lambda'} - E_{\lambda})}{2}\right) \\ &= 2 \sum_{\lambda, \lambda'} (p_{\lambda} - p_{\lambda'}) |A_{\lambda, \lambda'}|^2 \frac{p_{\lambda}/p_{\lambda'} - 1}{p_{\lambda}/p_{\lambda'} + 1} \\ &= 2 \sum_{\lambda, \lambda'} \frac{(p_{\lambda} - p_{\lambda'})^2}{p_{\lambda} + p_{\lambda'}} |A_{\lambda, \lambda'}|^2 \end{aligned} \quad (1.216)$$

where $p_{\lambda} = e^{-\beta E_{\lambda}}/Z$.

For unitary encoding $U(\theta) = e^{-i\theta H}$, $\rho_0 = \sum \lambda_k |k\rangle \langle k|$, we have $\rho(\theta) = U(\theta)\rho_0 U(\theta)$. Thus the Fisher Information reads

$$\mathcal{F}_Q[\rho_0; H] = 2 \sum_{k, k'} \frac{(\lambda_k - \lambda_{k'})^2}{\lambda_k + \lambda_{k'}} |\langle k| H |k'\rangle|^2 \quad (1.217)$$

Chapter 2

Form factor and Correlation function

2.1 The XXZ Heisenberg spin- $\frac{1}{2}$ inhomogeneous finite chain

The Hamiltonian is given by

$$H_{XXZ} = J \sum_{m=1} (\sigma_m^x \sigma_{m+1}^x + \sigma_m^y \sigma_{m+1}^y + \Delta (\sigma_m^z \sigma_{m+1}^z - 1)) \quad (2.1)$$

And $\Delta = 1$ corresponds to XXX chain.

The quantum R-matrix is given by

$$R(\lambda, \mu) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & b(\lambda, \mu) & c(\lambda, \mu) & 0 \\ 0 & c(\lambda, \mu) & b(\lambda, \mu) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.2)$$

where

$$b(\lambda, \mu) = \frac{\phi(\lambda - \mu)}{\phi(\lambda - \mu + \eta)} \quad (2.3)$$

$$c(\lambda, \mu) = \frac{\phi(\eta)}{\phi(\lambda - \mu + \eta)} \quad (2.4)$$

and

$$\Delta = \frac{1}{2}(q + q^{-1}) \quad q = e^\eta$$

with ϕ defined as

$$\phi(\lambda) = \begin{cases} \lambda & \text{XXX case} \\ \sinh(\lambda) & \text{XXZ case} \end{cases} \quad (2.5)$$

R-matrix satisfy Yang-Baxter equation

$$R_{12}(\lambda_1, \lambda_2) R_{13}(\lambda_1, \lambda_3) R_{23}(\lambda_2, \lambda_3) = R_{23}(\lambda_2, \lambda_3) R_{13}(\lambda_1, \lambda_3) R_{12}(\lambda_1, \lambda_2) \quad (2.6)$$

unitary condition

$$R_{12}(\lambda_1, \lambda_2) R_{21}(\lambda_2, \lambda_1) = 1 \quad (2.7)$$

and crossing symmetry relation

$$(\gamma \otimes 1) R_{12}(\lambda_1^s, \lambda_2) (\gamma \otimes 1) = R_{21}^t(\lambda_2, \lambda_1) \rho(\lambda_1, \lambda_2) \quad (2.8)$$

where

$$\gamma^2 = 1 \quad \gamma^t = \pm \gamma \quad (2.9)$$

and For the rational case

$$\lambda_1^s = \lambda_1 - \eta, \quad \gamma = \sigma^y, \quad \rho(\lambda_1, \lambda_2) = \frac{\lambda_1 - \lambda_2 - \eta}{\lambda_1 - \lambda_2} \quad (2.10)$$

and for the trigonometric case

$$\lambda_1^s = \lambda_1 - \eta + i\pi, \quad \gamma = \sigma^x, \quad \rho(\lambda_1, \lambda_2) = \frac{\sinh(\lambda_1 - \lambda_2 - \eta)}{\sinh(\lambda_1 - \lambda_2)} \quad (2.11)$$

We define the quantum L-operator acting on $C^2 \otimes \mathbb{H}_n$

$$L_n(\lambda, \xi_n) = R^{0n}(\lambda, \xi_n) \quad (2.12)$$

The quantum monodromy matrix is given by

$$T_0(\lambda) = T_{0,1,\dots,N}(\lambda; \xi_1, \dots, \xi_N) = R_{0N}(\lambda, \xi_N) \dots R_{01}(\lambda, \xi_1) = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix} \quad (2.13)$$

where $A(\lambda)$ acting on $\mathbb{H} = \otimes_{n=1}^N \mathbb{H}_n$.

$$R_{12}(\lambda, \mu) T_1(\lambda) T_2(\mu) = T_2(\mu) T_1(\lambda) R_{12}(\lambda, \mu) \quad (2.14)$$

with $T_1(\lambda) = T(\lambda) \otimes Id$ and $T_2(\mu) = Id \otimes T(\mu)$.

Moreover, the monodromy matrix satisfies the crossing symmetry relation

$$T_{0,1,\dots,N}(\lambda; \xi_1, \dots, \xi_N) \gamma_0 T_{0,1,\dots,N}^{t_0}(\lambda^s; \xi_1, \dots, \xi_N) \gamma_0 = \rho(\lambda; \xi_1, \dots, \xi_N) 1 \quad (2.15)$$

where $\rho(\lambda; \xi_1 \xi_N) = \prod_{i=1}^N \rho(\lambda, \xi_i)$ and γ_0 is matrix γ .

The transfer matrix $\mathcal{T}(\lambda)$ is defined as the trace of the total monodromy matrix

$$\mathcal{T}(\lambda) = A(\lambda) + D(\lambda) \quad (2.16)$$

Suppose a reference state $|0\rangle$ called psedo-vacuum

$$\begin{aligned} A(\lambda) |0\rangle &= a(\lambda) |0\rangle \\ D(\lambda) |0\rangle &= d(\lambda) |0\rangle \\ C(\lambda) |0\rangle &= 0 \\ B(\lambda) |0\rangle &\neq 0 \end{aligned} \quad (2.17)$$

where $a(\lambda) = 1, d(\lambda) = \prod_{i=1}^N b(\lambda, \xi_i)$ For XXX or XXZ model $|0\rangle$ is the state all spins up.

We would like to find the common eigenstates of the transfer matrices for different spectral parameter λ .

The Bethe equations

$$r(\lambda_k) \prod_{j=1, j \neq k}^n \frac{b(\lambda_k, \lambda_j)}{b(\lambda_j, \lambda_k)} = 1, \quad 1 \leq k \leq n \quad (2.18)$$

with

$$r(\lambda) = \frac{a(\lambda)}{d(\lambda)} \quad (2.19)$$

The common eigenstates is obtained by

$$\prod_{j=1}^n B(\lambda_j) |0\rangle \quad (2.20)$$

The corresponding eigenvalue is

$$\tau(\mu, \{\lambda_j\}) = a(\mu) \prod_{j=1}^n b^{-1}(\lambda_j, \mu) + d(\mu) \prod_{j=1}^n b^{-1}(\mu, \lambda_j) \quad (2.21)$$

We give the R-matrix $R_{1\dots n}^\sigma(\xi_1, \dots, \xi_n)$ constructed as ordered product of elementary R-matrices $R_{ij}(\xi_i, \xi_j)$, such that

$$R_{1\dots n}^\sigma(\xi_1, \dots, \xi_n) T_{0,1\dots n}(\lambda; \xi_1, \dots, \xi_n) = T_{0,\sigma(1),\dots,\sigma(n)}(\lambda; \xi_{\sigma(1)}, \dots, \xi_{\sigma(n)}) R_{1,\dots,n}^\sigma(\xi_1, \dots, \xi_n) \quad (2.22)$$

For any element σ of S_n , factorizing F-matrix satisfies

$$F_{\sigma(1),\dots,\sigma(n)}(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)}) R_{1,\dots,n}^\sigma(\xi_1, \dots, \xi_n) = F_{1,\dots,n}(\xi_1, \dots, \xi_n) \quad (2.23)$$

Factorizing F-matrices can be given as an ordered product of partial F-matrices

$$F_{1\dots n} = F_{2\dots n} F_{1,2\dots n} = \dots = F_{n-1n} F_{n-2,n-1n} \dots F_{1,2\dots n} \quad (2.24)$$

The partial F-matrices $F_{i,i+1\dots n} = F_{i,i+1\dots n}(\xi_i; \xi_{i+1}, \dots, \xi_n)$ are given in terms of partial R-matrices

$$F_{i,i+1\dots n}(\xi_i; \xi_{i+1}, \dots, \xi_n) = e_i^{(11)} + e_i^{(22)} R_{i,i+1\dots n}(\xi_i; \xi_{i+1}, \dots, \xi_n) \quad (2.25)$$

where the partial R-matrices acting in $V_i \otimes \dots \otimes V_n$

$$R_{i,i+1\dots n}(\xi_i; \xi_{i+1}, \dots, \xi_n) = R_{in}(\xi_i, \xi_n) \dots R_{i,i+1}(\xi_i, \xi_{i+1}) \quad (2.26)$$

The partial F-matrix $F_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N)$ has a useful expression

$$F_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N) = \begin{pmatrix} 1 & 0 \\ C_{1\dots N}(\lambda; \xi_1, \xi_2, \dots, \xi_N) & D_{1,\dots,N}(\lambda; \xi_1, \xi_2, \dots, \xi_N) \end{pmatrix} \quad (2.27)$$

Moreover, the relations above lead to

$$F_{1\dots N}(\xi_1, \dots, \xi_N) T_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N) F_{1\dots N}^{-1}(\xi_1, \dots, \xi_N) = F_{\sigma(1)\dots\sigma(N)}(\xi_{\sigma(1)}, \dots, \xi_{\sigma(N)}) T_{0,\sigma(1),\dots,\sigma(N)}(\lambda; \xi_{\sigma(1)}, \dots, \xi_{\sigma(N)}) F_{\sigma(1),\dots,\sigma(N)}^{-1}(\xi_{\sigma(1)}, \dots, \xi_{\sigma(N)}) \quad (2.28)$$

which means that the monodromy matrix $\tilde{T}$ in the F-basis

$$\tilde{T}_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N) = F_{1\dots N}(\xi_1, \dots, \xi_N) T_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N) F_{1\dots N}^{-1}(\xi_1, \dots, \xi_N) \quad (2.29)$$

is totally symmetric.

The element of monodromy matrix is simple in the F-basis

$$\tilde{D}_{1\dots N}(\lambda; \xi_1, \dots, \xi_N) = F_{1,\dots,N}(\xi_1, \dots, \xi_N) D_{1,\dots,N}(\lambda; \xi_1, \dots, \xi_N) F_{1,\dots,N}^{-1}(\xi_1, \dots, \xi_N) = \otimes_{i=1}^N \begin{pmatrix} b(\lambda, \xi_i) & 0 \\ 0 & 1 \end{pmatrix} \quad (2.30)$$

$$\tilde{B}_{1\dots N}(\lambda) = \sum_{i=1}^N \sigma_i^- c(\lambda, \xi_i) \otimes_{j \neq i} \begin{pmatrix} b(\lambda, \xi_j) & 0 \\ 0 & b^{-1}(\xi_j, \xi_i) \end{pmatrix} \quad (2.31)$$

$$\tilde{C}_{1\dots N}(\lambda) = \sum_{i=1}^N \sigma_i^+ c(\lambda, \xi_i) \otimes_{j \neq i} \begin{pmatrix} b(\lambda, \xi_j) b^{-1}(\xi_i, \xi_j) & 0 \\ 0 & 1 \end{pmatrix} \quad (2.32)$$

and $\tilde{A}$ can be obtained from quantum determinant relations (2.15).

It can be shown that the psedo-vacuum state is left invariant under such a basis change i.e. $F|0\rangle = |0\rangle$. Thus for scalar products of the quantum states

$$\langle 0| C(\lambda_1) \dots C(\lambda_n) B(\lambda_{n+1}) \dots B(\lambda_{2n}) |0\rangle = \langle 0| \tilde{C}(\lambda_1) \tilde{C}(\lambda_2) \dots \tilde{B}(\lambda_{n+1}) \dots \tilde{B}(\lambda_{2n}) |0\rangle \quad (2.33)$$

We have the expression for F-matrices $F_{1\dots n}$ in the F-basis

$$F_{1\dots n} = \tilde{F}_{1,2\dots n} F_{2\dots n} = \dots = \tilde{F}_{1,2\dots n} \tilde{F}_{2,3\dots n} \dots F_{n-1n} \quad (2.34)$$

where

$$\begin{aligned} \tilde{F}_{1,2\dots n}(\xi_1; \xi_2, \dots, \xi_n) &= F_{2\dots n} F_{1,2\dots n}(\xi_1; \xi_2, \dots, \xi_n) F_{2\dots n}^{-1}(\xi_2, \dots, \xi_n) \\ &= \begin{pmatrix} 1 & 0 \\ \tilde{C}_{2\dots n}(\xi_1; \xi_2, \dots, \xi_n) & \tilde{D}_{2\dots n}(\xi_1; \xi_2, \dots, \xi_n) \end{pmatrix} \end{aligned} \quad (2.35)$$

2.2 Scalar products of Bethe states and the Gaudin formula

We calculate

$$S_n(\{\mu_j\}, \{\lambda_k\}) = \langle 0 | \prod_{j=1}^n C(\mu_j) \prod_{k=1}^n B(\lambda_k) | 0 \rangle \quad (2.36)$$

$\{\lambda_k\}$ is a solution of Bethe equations, then

$$(A(\mu) + D(\mu)) \prod_{k=1}^n B(\lambda_k) | 0 \rangle = \tau(\mu, \{\lambda_k\}) \prod_{k=1}^n B(\lambda_k) | 0 \rangle \quad (2.37)$$

with the eigenvalue

$$\tau(\mu, \{\lambda_j\}) = a(\mu) \prod_{j=1}^n b^{-1}(\lambda_j, \mu) + d(\mu) \prod_{j=1}^n b^{-1}(\mu, \lambda_j) \quad (2.38)$$

Theorem 2.2.1. *Let $\{\lambda_1, \dots, \lambda_n\}$ be a solution of Bethe equations*

$$\frac{d(\lambda_j)}{a(\lambda_j)} \prod_{k \neq j} \frac{b(\lambda_k, \lambda_j)}{b(\lambda_j, \lambda_k)} = 1, \quad 1 \leq j \leq n$$

and $\{\mu_1, \dots, \mu_n\}$ be an arbitrary set of parameters. Then the scalar product (2.36) can be represented as a ratio of two determinants

$$S_n(\{\mu_j\}, \{\lambda_k\}) = S_n(\{\lambda_k\}, \{\mu_j\}) = \frac{\det \mathbf{T}(\{\mu_j\}, \{\lambda_k\})}{\det \mathbf{V}(\{\mu_j\}, \{\lambda_k\})} \quad (2.39)$$

where T and V are $n \times n$ matrices defined as

$$T_{ab} = \frac{\partial}{\partial \lambda_a} \tau(\mu_b, \{\lambda_k\}), \quad V_{ab} = \frac{1}{\phi(\mu_b - \lambda_a)}, \quad 1 \leq a, b \leq n \quad (2.40)$$

Proof: See primary paper.

2.3 Quantum inverse scattering problem for local spins and correlation functions

2.3.1 Local spin operators in the F-basis and quantum inverse problem

The spin operators in F-basis at the first or the last site of the chain can be easily obtained due to the very simple form (2.35) of $\tilde{F}_{1,2\dots N}$

$$F_{1\dots N}\sigma_1^-F_{1\dots N}^{-1} = \tilde{F}_{1,2\dots N}F_{2\dots N}\sigma_1^-F_{2\dots N}^{-1}\tilde{F}_{1,2\dots N}^{-1} = \tilde{D}_{1\dots N}(\xi_1;\xi_1,\xi_2,\dots,\xi_N)\sigma_1^- \quad (2.41)$$

and by symmetry so do the last site spin operator.

Taking into account that $\tilde{D}$ is totally symmetric, for operator on site i we obtain:

$$F_{i\dots N1\dots i-1}\sigma_i^-F_{i\dots N1\dots i-1}^{-1} = \tilde{D}_{1\dots N}(\xi_i)\sigma_i^- \quad (2.42)$$

Thus our next goal is to evaluate the product of two "permuted" F-matrices $F_{1\dots N}F_{i\dots N1\dots i-1}^{-1}$, which can be considered as the expression in the F-basis of a propagator $F_{i\dots N1\dots i-1}^{-1}F_{1\dots N}$.

Theorem 2.3.1. *Let U_1^i be the propagator $F_{i\dots N1\dots i-1}^{-1}F_{1\dots N}$ from site 1 to site i of the chain. It can be written into the following forms:*

$$U_1^i = R_{i-1,i\dots N1\dots i-2\dots R_{2,3\dots N1}R_{1,2\dots N} \quad (2.43)$$

$$= \prod_{\alpha=1}^{i-1} (A_{1\dots N}(\xi_\alpha) + D_{1\dots N}(\xi_\alpha)) \quad (2.44)$$

Note that in (2.44), all factors commute with each other.

Remark 2.3.2. The propagator U_1^1 through the whole chain being the identity, note that

$$\prod_{\alpha=1}^N (A_{1\dots N}(\xi_\alpha) + D_{1\dots N}(\xi_\alpha)) = R_{N,1\dots N-1\dots R_{1,2\dots N} = 1 \quad (2.45)$$

Thus the inverse $(U_1^i)^{-1}$ of the propagator on a part of the chain is merely the propagator $U_i^1 = \prod_{\alpha=i}^N (A + D)(\xi_\alpha)$ on the remaining part.

Remark 2.3.3. The action of propagator U_1^i on an operator entry $X_{1\dots N}$ of the monodromy matrix ($X = A, B, C, D$):

$$U_1^i X_{1\dots N} = X_{i\dots N1\dots i-1} U_1^i \quad (2.46)$$

or equivalently

$$U_1^i T_{0,1\dots N} = T_{0,i\dots N1\dots i-1} U_1^i \quad (2.47)$$

Theorem 2.3.1 allows us to obtain the value of any local operator in F-basis:

$$F_{1\dots N}\sigma_i^-F_{1\dots N}^{-1} = \prod_{\alpha=1}^{i-1} (\tilde{A} + \tilde{D})(\xi_\alpha) \cdot \tilde{D}(\xi_i)\sigma_i^- \cdot \prod_{\alpha=1}^{i-1} (\tilde{A} + \tilde{D})^{-1}(\xi_\alpha) \quad (2.48)$$

Note that $\tilde{B}$ and $\tilde{C}$ are quasilocal in terms of the σ_i^- or σ_i^+ . Indeed a direct calculus in the F-basis gives the identity:

$$\tilde{D}(\xi_i)\sigma_i^-(\tilde{A} + \tilde{D})(\xi_i) = \tilde{B}(\xi_i) \quad (2.49)$$

The proof of the last formula is as below.

2.3.2 Reconstruction of local spin operators

Theorem 2.3.4. *Local spin operators at a given site i of the inhomogeneous XXX or XXZ Heisenberg chain are given by:*

$$\sigma_i^- = \prod_{\alpha=1}^{i-1} (A+D)(\xi_\alpha) \cdot B(\xi_i) \cdot \prod_{\alpha=i+1}^N (A+D)(\xi_\alpha) \quad (2.50)$$

$$\sigma_i^+ = \prod_{\alpha=1}^{i-1} (A+D)(\xi_\alpha) \cdot C(\xi_i) \cdot \prod_{\alpha=i+1}^N (A+D)(\xi_\alpha) \quad (2.51)$$

$$\sigma_i^z = \prod_{\alpha=1}^{i-1} (A+D)(\xi_\alpha) \cdot (A-D)(\xi_i) \cdot \prod_{\alpha=i+1}^N (A+D)(\xi_\alpha) \quad (2.52)$$

Proof is by taking x_0 with $\sigma_0^-, \sigma_0^+, \sigma_0^z$ respectively for the Lemma below

Lemma 2.3.5. *Let x_i act on quantum spin space V_i and x_0 on auxiliary space V_0 . They are related by the identity:*

$$\text{tr}_0(x_0 R_{0,1\dots N}(\xi_i)) = \prod_{\alpha=1}^{i-1} (A+D)^{-1}(\xi_\alpha) \cdot x_i \cdot \prod_{\alpha=1}^i (A+D)(\xi_\alpha) \quad (2.53)$$

2.3.3 General formula for correlation functions

Here we give the k -point spin-spin correlation function between two Bethe states. Indeed, for any integer k and any subset $\{i_j\}_{1 \leq j \leq k}$ of $\{1 \dots N\}$, with the convention $i_1 \leq i_2 \leq i_3 \dots \leq i_k$, the correlation function for spins at sites $i_1, \dots, i_k$ between two Bethe states $\langle 0 | C(\mu_1) \dots C(\mu_{n_1})$ and $B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle$ can be written into the following form:

$$\begin{aligned} & \langle 0 | C(\mu_1) \dots C(\mu_{n_1}) \sigma_{i_1}^{\epsilon_1} \sigma_{i_2}^{\epsilon_2} \dots \sigma_{i_k}^{\epsilon_k} B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle \\ &= \prod_{\alpha=1}^{i_1-1} \prod_{j=1}^{n_1} b^{-1}(\mu_j, \xi_\alpha) \cdot \prod_{\alpha=i_k+1}^N \prod_{j=1}^{n_2} b^{-1}(\lambda_j, \xi_\alpha) \\ & \times \langle 0 | C(\mu_1) \dots C(\mu_{n_1}) \cdot X^{\epsilon_1}(\xi_{i_1}) \cdot \prod_{\alpha=i_1+1}^{i_2-1} (A+D)(\xi_\alpha) \cdot X^{\epsilon_2}(\xi_{i_2}) \dots \\ & \times \dots \prod_{\alpha=i_k+1}^{i_k-1} (A+D)(\xi_\alpha) \cdot X^{\epsilon_k}(\xi_{i_k}) \cdot B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle \end{aligned} \quad (2.54)$$

where $\epsilon_i, 1 \leq j \leq k$ takes the values $+, -, \text{ or } z$, X^{ϵ_j} being respectively to $C, B, A-D$.

2.4 Form factors

2.4.1 operators σ_m^- and σ_m^+

We begin with two one-point functions:

$$F_n^-(m, \{\mu_j\}, \{\lambda_k\}) = \langle 0 | \prod_{j=1}^{n+1} C(\mu_j) \sigma_m^- \prod_{k=1}^n B(\lambda_k) | 0 \rangle \quad (2.55)$$

and

$$F_n^+(m, \{\lambda_k\}, \{\mu_j\}) = \langle 0 | \prod_{k=1}^n C(\lambda_k) \sigma_m^+ \prod_{j=1}^{n+1} B(\mu_j) | 0 \rangle \quad (2.56)$$

where $\{\lambda_k\}_n$ and $\{\mu_j\}_{n+1}$ are solutions of the Bethe equations.

Proposition 2.4.1. *For two Bethe states with spectral parameters $\{\lambda_k\}_n$ and $\{\mu_j\}_{n+1}$ the matrix element of the operator σ_m^- can be represented as a determinant,*

$$\begin{aligned} F_n^-(m, \{\mu_j\}, \{\lambda_k\}) &= \frac{\phi_{m-1}(\{\mu_j\}) \prod_{j=1}^{n+1} \varphi(\mu_j - \xi_m + \eta)}{\phi_{m-1}(\{\lambda_k\}) \prod_{k=1}^n \varphi(\lambda_k - \xi_m + \eta)} \\ &\quad \times \frac{1}{\prod_{n+1 \geq k > j \geq 1} \varphi(\mu_k - \mu_j) \prod_{1 \leq \beta \leq \alpha \leq n} \varphi(\lambda_\beta - \lambda_\alpha)} \\ &\quad \times \det_{n+1} H^-(m, \{\mu_j\}, \{\lambda_k\}) \end{aligned} \quad (2.57)$$

where the coefficients $\phi_m(\{\lambda_k\})$ are

$$\phi_m(\{\lambda_k\}) = \prod_{k=1}^n \prod_{j=1}^m b^{-1}(\lambda_k, \xi_j) \quad (2.58)$$

and the $(n+1) \times (n+1)$ matrix H^- is defined as

$$H_{ab}^-(m) = \frac{\varphi(\eta)}{\varphi(\mu_a - \lambda_b)} \left(a(\lambda_b) \prod_{\substack{j=1 \\ j \neq a}}^{n+1} \varphi(\mu_j - \lambda_b + \eta) - d(\lambda_b) \prod_{\substack{j=1 \\ j \neq a}}^{n+1} \varphi(\mu_j - \lambda_b - \eta) \right) \quad (2.59)$$

for $b < n+1$

$$H_{an+1}^-(m) = \frac{\varphi(\eta)}{\varphi(\mu_a - \xi_m + \eta) \varphi(\mu_a - \xi_m)} \quad (2.60)$$

And $F_n^+(m, \{\lambda_k\}, \{\mu_j\})$ share a similar representation,

$$F_n^+(m, \{\lambda_k\}, \{\mu_j\}) = \frac{\phi_m(\lambda_k) \phi_{m-1}(\lambda_k)}{\phi_{m-1}(\mu_j) \phi_m(\mu_j)} F_n^-(m, \{\lambda_k\}, \{\mu_j\}) \quad (2.61)$$

2.4.2 Operator σ_m^z

We calculate

$$F_n^z(m, \{\mu_j\}, \{\lambda_k\}) = \langle 0 | \prod_{j=1}^n C(\mu_j) \sigma_m^z \prod_{k=1}^n B(\lambda_k) | 0 \rangle \quad (2.62)$$

Proposition 2.4.2. *The one-point function can be represented as a determinant:*

$$\begin{aligned} F_n^z(m, \{\mu_j\}, \{\lambda_k\}) &= \frac{\phi_{m-1}(\{\mu_j\})}{\phi_{m-1}(\{\lambda_k\})} \prod_{j=1}^n \frac{\varphi(\mu_j - \xi_m + \eta)}{\varphi(\lambda_j - \xi_m + \eta)} \\ &\quad \times \frac{1}{\prod_{j>k} \varphi(\mu_k - \mu_j) \prod_{\alpha<\beta} \varphi(\lambda_\beta - \lambda_\alpha)} \\ &\quad \times \det_n (H(\{\mu_j\}, \{\lambda_k\}) - 2P(m, \{\mu_j\}, \{\lambda_k\})) \end{aligned} \quad (2.63)$$

where

$$\begin{aligned} H_{ab} &= \frac{\varphi(\eta)}{\varphi(\mu_a - \lambda_b)} \left(a(\lambda_b) \prod_{j \neq a} \varphi(\mu_j - \lambda_b + \eta) - d(\lambda_b) \prod_{j \neq a} \varphi(\mu_j - \lambda_b - \eta) \right) \\ 1 &\leq a, b \leq n \end{aligned} \quad (2.64)$$

2.5 The XXZ spin- $\frac{1}{2}$ Heisenberg chain

We give the XXZ Heisenberg *spin* $-\frac{1}{2}$ finite chain of length M

$$H_{XXZ} = \sum_{m=1}^M \left(\sigma_m^x \sigma_{m+1}^x + \sigma_m^y \sigma_{m+1}^y + \Delta (\sigma_m^z \sigma_{m+1}^z - 1) - \frac{h}{2} \sigma_m^z \right) \quad (2.65)$$

We can see the ground state is $\begin{cases} \Delta \leq -1 & \text{ferromagnetic} \\ \Delta > -1 & \text{anti-ferromagnetic} \end{cases}$

We focus on the second condition and decompose this domain into a massive regime (for $\Delta > 1$) and a gapless regime (for $-1 < \Delta < 1$) in the thermodynamic limit ($M \rightarrow \infty$)

The R-matrix of the XXZ model is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & b(\lambda, \mu) & c(\lambda, \mu) & 0 \\ 0 & c(\lambda, \mu) & b(\lambda, \mu) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.66)$$

where

$$b(\lambda, \mu) = \frac{\sinh(\lambda - \mu)}{\sinh(\lambda - \mu + \eta)} \quad c(\lambda, \mu) = \frac{\sinh \eta}{\sinh(\lambda - \mu + \eta)} \quad (2.67)$$

where

$$\Delta = \frac{1}{2}(q + q^{-1}) \quad q = e^\eta$$

Here the R-matrix act on space $V_1 \otimes V_2$ and depends generically on two spectral parameters λ_1 and λ_2 . We define the *quantum L-operator* acting on space $V_0 \otimes \mathbb{H}_m$

$$L_m(\lambda, \xi_m) = R_{0m}(\lambda - \xi_m) \quad (2.68)$$

The monodromy matrix is constructed as an ordered product of such L-operators

$$T(\lambda) = R_{0M}(\lambda - \xi_M) \cdots R_{02}(\lambda - \xi_2) R_{01}(\lambda - \xi_1) = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix} \quad (2.69)$$

Transfer matrices commute with each other for different values of the spectral parameter λ . They also commute with the Hamiltonian. The Hamiltonian can be reconstructed in terms of the transfer matrix by means of the following "trace identity"

$$H_{XXZ} = 2 \sinh \eta \frac{\partial}{\partial \lambda} \log \mathcal{T}(\lambda)|_{\lambda=\xi_j} + \text{const} \quad (2.70)$$

Common eigenstates of transfer matrix and Hamiltonian are obtained by $B(\lambda_1) \dots B(\lambda_N) |0\rangle$, where $\{\lambda_k\}$ is a solution of Bethe equations,

$$\frac{a(\lambda_j)}{d(\lambda_j)} \prod_{\substack{k=1 \\ k \neq j}}^N \frac{b(\lambda_j, \lambda_k)}{b(\lambda_k, \lambda_j)} = 1, \quad 1 \leq j \leq N \quad (2.71)$$

where $a(\lambda) = 1$ and $d(\lambda) = \prod_{i=1}^M b(\lambda, \xi_i)$. The corresponding eigenvalue for the transfer matrix $\mathcal{T}(\mu)$ is then

$$\tau(\mu, \{\lambda_j\}) = a(\mu) \prod_{j=1}^n b^{-1}(\lambda_j, \mu) + d(\mu) \prod_{j=1}^n b^{-1}(\mu, \lambda_j) \quad (2.72)$$

The logarithmic form of Bethe equations:

$$M p_{0_{tot}}(\lambda_j) + \sum_{k=1}^N \theta(\lambda_j - \lambda_k) = 2\pi n_j, \quad 1 \leq j \leq N, \quad (2.73)$$

where n_j are integers for N odd and half integer for N even. And

$$p_{0_{tot}}(\lambda) = \frac{i}{M} \ln \frac{d(\lambda)}{a(\lambda)} = \frac{1}{M} \sum_{k=1}^M p_0(\lambda - \xi_k + \eta/2) \quad (2.74)$$

$$p_0(\lambda) = i \ln \frac{\sinh(\lambda - \frac{\eta}{2})}{\sinh(\lambda + \frac{\eta}{2})} \quad (2.75)$$

$$\theta(\lambda) = i \ln \frac{\sinh(\eta + \lambda)}{\sinh(\eta - \lambda)} \quad (2.76)$$

In the thermodynamic limit ($M \rightarrow \infty$), we give the Lieb equation for the ground state:

$$\rho_{tot}(\alpha) + \int_{-\Lambda}^{\Lambda} K(\alpha - \beta) \rho_{tot}(\beta) d\beta = \frac{p'_{0_{tot}}(\alpha)}{2\pi} \quad (2.77)$$

where

$$\begin{aligned} \alpha &= \lambda & \text{for } -1 < \Delta < 1 \\ \alpha &= i\lambda & \text{for } \Delta > 1 \end{aligned} \quad (2.78)$$

And

$$\begin{aligned} K(\alpha) &= \frac{\partial}{\partial \alpha} (-\theta(\lambda(\alpha)) / 2\pi) = \frac{\sin 2\zeta}{2\pi \sinh(\alpha + i\zeta) \sinh(\alpha - i\zeta)} \\ p_0(\alpha) &= \frac{\partial}{\partial \alpha} p_{0_{tot}}(\lambda(\alpha)) = \frac{\sin \zeta}{\sinh\left(\alpha + i\frac{\zeta}{2}\right) \sinh(\alpha - i\zeta/2)} \end{aligned} \quad (2.79)$$

for $-1 < \Delta < 1$, with $\zeta = i\eta$ And

$$\begin{aligned} K(\alpha) &= \frac{\partial}{\partial \alpha} (-\theta(\lambda(\alpha)) / 2\pi) = \frac{\sinh 2\zeta}{2\pi \sin(\alpha + i\zeta) \sin(\alpha - i\zeta)} \\ p'_0(\alpha) &= \frac{\partial}{\partial \alpha} p_{0_{tot}}(\lambda(\alpha)) = \frac{\sinh \zeta}{\sin\left(\alpha + i\frac{\zeta}{2}\right) \sin\left(\alpha - i\frac{\zeta}{2}\right)} \end{aligned} \quad (2.80)$$

for $\Delta > 1$, with $\zeta = -\eta$, with

$$p'_{0_{tot}}(\alpha) = \frac{1}{M} \sum_{i=1}^M p'_0(\alpha - \beta_k - i\frac{\zeta}{2}) \quad (2.81)$$

where $\beta_k = \xi_k$ in the domain $-1 < \Delta < 1$, and $\beta_k = i\xi_k$ in the domain $\Delta > 1$. The integration limit Λ is equal to $\frac{\pi}{2}$ and to ∞ for $-1 < \Delta < 1$. In the homogeneous model where all parameter ξ_k are equal to $\frac{\eta}{2}$, the density for the ground state is given by

$$\rho(\alpha) = \frac{1}{2\zeta \cosh\left(\frac{\pi\alpha}{\zeta}\right)} \quad \text{for } -1 < \Delta < 1 \quad (2.82)$$

$$\rho(\alpha) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \frac{e^{2in\alpha}}{\cosh(n\zeta)} \quad \text{for } \Delta > 1 \quad (2.83)$$

And the solution of inhomogeneous model is given by

$$\rho_{tot}(\alpha) = \frac{1}{M} \sum_{i=1}^M \rho\left(\alpha - \beta_k - i\frac{\zeta}{2}\right) \quad (2.84)$$

where the inhomogeneous parameter are contained in the region $-\zeta < \text{Im}\beta_j < 0$.

2.6 Quantum inverse scattering problem and correlation functions

We want to compute the general matrix element of products of local spin operators between two Bethe states $\langle 0 | \prod_j C(\mu_j)$ and $\prod_k B(\lambda_k) | 0 \rangle$.

$$\langle 0 | \prod_j C(\mu_j) \left(\prod_i \sigma_i^{\epsilon_i} \right) \prod_k B(\lambda_k) | 0 \rangle \quad (2.85)$$

where $\sigma_i^{\epsilon_i}, \epsilon_i \in \{+, -, z\}$ are pauli spin operators at site i .

2.6.1 Solution of the quantum inverse problem

Theorem 2.6.1. *Local spin operators at a given site i of the inhomogeneous XXX or XXZ Heisenberg chain are given by:*

$$\sigma_i^- = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot B(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (2.86)$$

$$\sigma_i^+ = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot C(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (2.87)$$

$$\sigma_i^z = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot (A - D)(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (2.88)$$

Proof is by taking x_0 with $\sigma_0^-, \sigma_0^+, \sigma_0^z$ respectively for the Lemma below

Here we give the k-point spin-spin correlation function between two Bethe states. Indeed, for any integer k and any subset $\{i_j\}_{1 \leq j \leq k}$ of $\{1 \dots N\}$, with the convention $i_1 \leq i_2 \leq i_3 \dots \leq i_k$, the correlation function for spins at sites $i_1, \dots, i_k$ between two Bethe states $\langle 0 | C(\mu_1) \dots C(\mu_{n_1})$ and $B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle$ can be written into the following form:

$$\begin{aligned}
& \langle 0 | C(\mu_1) \dots C(\mu_{n_1}) \sigma_{i_1}^{\epsilon_1} \sigma_{i_2}^{\epsilon_2} \dots \sigma_{i_k}^{\epsilon_k} B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle \\
&= \prod_{\alpha=1}^{i_1-1} \prod_{j=1}^{n_1} b^{-1}(\mu_j, \xi_\alpha) \cdot \prod_{\alpha=i_k+1}^N \prod_{j=1}^{n_2} b^{-1}(\lambda_j, \xi_\alpha) \\
&\times \langle 0 | C(\mu_1) \dots C(\mu_{n_1}) \cdot X^{\epsilon_1}(\xi_{i_1}) \cdot \prod_{\alpha=i_1+1}^{i_2-1} (A+D)(\xi_\alpha) \cdot X^{\epsilon_2}(\xi_{i_2}) \dots \\
&\times \dots \prod_{\alpha=i_k+1}^{i_k-1} (A+D)(\xi_\alpha) \cdot X^{\epsilon_k}(\xi_{i_k}) \cdot B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle
\end{aligned} \tag{2.89}$$

where $\epsilon_i, 1 \leq j \leq k$ takes the values $+, -, \text{ or } z$, X^{ϵ_j} being respectively to $C, B, A - D$.

2.6.2 Action of operators A, B, C, D on a general state

$$\langle 0 | \prod_{k=1}^N C(\lambda_k) A(\lambda_{N+1}) = \sum_{a'=1}^{N+1} a(\lambda_{a'}) \frac{\prod_{k=1}^N \sinh(\lambda_k - \lambda_{a'} + \eta)}{\prod_{\substack{k=1 \\ k \neq a'}}^{N+1} \sinh(\lambda_k - \lambda_{a'})} \langle 0 | \prod_{\substack{k=1 \\ k \neq a'}}^{N+1} C(\lambda_k) \tag{2.90}$$

$$\langle 0 | \prod_{k=1}^N D(\lambda_{N+1}) = \sum_{a=1}^{N+1} d(\lambda_a) \frac{\prod_{k=1}^N \sinh(\lambda_a - \lambda_k + \eta)}{\prod_{\substack{k=1 \\ k \neq a}}^{N+1} \sinh(\lambda_a - \lambda_k)} \langle 0 | \prod_{\substack{k=1 \\ k \neq a}}^{N+1} C(\lambda_k) \tag{2.91}$$

2.6.3 Scalar products

Theorem 2.6.2. *Let $\{\lambda_1, \dots, \lambda_n\}$ be a solution of Bethe equations*

$$\frac{d(\lambda_j)}{a(\lambda_j)} \prod_{k \neq j} \frac{b(\lambda_k, \lambda_j)}{b(\lambda_j, \lambda_k)} = 1, \quad 1 \leq j \leq n$$

and $\{\mu_1, \dots, \mu_n\}$ be an arbitrary set of parameters. Then the scalar product

$$S_n(\{\mu_j\}, \{\lambda_k\}) = \langle 0 | \prod_{j=1}^n C(\mu_j) \prod_{k=1}^n B(\lambda_k) | 0 \rangle \tag{2.92}$$

can be represented as a ratio of two determinants

$$S_n(\{\mu_j\}, \{\lambda_k\}) = S_n(\{\lambda_k\}, \{\mu_j\}) = \frac{\det \mathbf{T}(\{\mu_j\}, \{\lambda_k\})}{\det \mathbf{V}(\{\mu_j\}, \{\lambda_k\})} \tag{2.93}$$

where T and V are $n \times n$ matrices defined as

$$T_{ab} = \frac{\partial}{\partial \lambda_a} \tau(\mu_b, \{\lambda_k\}), \quad V_{ab} = \frac{1}{\phi(\mu_b - \lambda_a)}, \quad 1 \leq a, b \leq n \tag{2.94}$$

When the two states are equal, one obtain the Gaudin formula for the norm of a Bethe state

$$\langle 0 | \prod_{j=1}^N C(\lambda_j) \prod_{k=1}^N B(\lambda_k) | 0 \rangle = \sinh^N \eta \prod_{\alpha \neq \beta} \frac{\sinh(\lambda_\alpha - \lambda_\beta + \eta)}{\sinh(\lambda_\alpha - \lambda_\beta)} \det \Phi'(\{\lambda_\alpha\}) \quad (2.95)$$

where Φ' is a $N \times N$ matrix whose elements are given by

$$\Phi'_{ab} = -\frac{\partial}{\partial \lambda_b} \ln \left(\frac{a(\lambda_a)}{d(\lambda_a)} \prod_{\substack{k=1 \\ k \neq a}}^N \frac{b(\lambda_a, \lambda_k)}{b(\lambda_k, \lambda_a)} \right) \quad (2.96)$$

2.6.4 Thermodynamic limit

We have the property

$$\frac{1}{M} \sum_{j=1}^N f(\alpha_j) = \int_{-\Lambda}^{\Lambda} f(\alpha) \rho_{tot}(\alpha) d\alpha + O(M^{-\infty}) \quad (2.97)$$

The Gaudin matrix in the thermodynamic limit

$$\Phi'_{ab}(\alpha) = -2i\pi M \left(\delta_{ab} \rho_{tot}(\alpha_a) + \frac{1}{M} K(\alpha_a - \alpha_b) \right) + O(M^{-\infty}) \quad (2.98)$$

$$\text{for } -1 < \Delta < 1 \quad (2.99)$$

$$\Phi'_{ab}(\alpha) = 2\pi M \left(\delta_{ab} \rho_{tot}(\alpha_a) + \frac{1}{M} K(\alpha_a - \alpha_b) \right) + O(M^{-\infty}) \quad (2.100)$$

$$\text{for } \Delta > 1 \quad (2.101)$$

2.7 Emptiness formation probability

The emptiness formation probability is the probability to detect a ferromagnetic domain of length m in the antiferromagnetic ground state:

$$\tau(m) = \frac{\langle \psi_g | \prod_{j=1}^m \frac{1}{2} (1 - \sigma_j^z) | \psi_g \rangle}{\langle \psi_g | \psi_g \rangle} \quad (2.102)$$

We have for the operator $\frac{1}{2}(1 - \sigma_j^z)$ by the quantum inverse scattering problem

$$\frac{1}{2}(1 - \sigma_j^z) = \prod_{k=1}^{j-1} (A(\xi_k) + D(\xi_k)) D(\xi_j) \prod_{k=j+1}^M (A(\xi_k) + D(\xi_k)) \quad (2.103)$$

So

$$\tau(m) = \phi(\{\lambda\}) \frac{\langle 0 | \prod_{a=1}^N C(\lambda_a) \prod_{j=1}^m D(\xi_j) \prod_{a=1}^N B(\lambda_a) | 0 \rangle}{\langle 0 | \prod_{a=1}^N C(\lambda_a) \prod_{a=1}^N B(\lambda_a) | 0 \rangle} \quad (2.104)$$

where $\phi_m(\{\lambda\})$ is eigenvalue of tranfer matrix

$$\phi_m(\{\lambda\}) = \prod_{j=1}^m \prod_{a=1}^N \frac{\sinh(\lambda_a - \xi_j)}{\sinh(\lambda_a - \xi_j + \eta)} \quad (2.105)$$

Using relation (2.91) we get

$$\begin{aligned} \langle 0 | \prod_{k=1}^N C(\lambda_k) \prod_{j=1}^m D(\lambda_{N+j}) &= \sum_{a_1=1}^{N+1} \sum_{\substack{a_2=1 \\ a_2 \neq a_1}}^{N+2} \dots \\ &\times \sum_{\substack{a_m=1 \\ a_m \neq a_1 \dots a_{m-1}}}^{N+m} G_{a_1 \dots a_m}(\lambda_1 \dots \lambda_{N+m}) \\ &\times \langle 0 | \prod_{\substack{k=1 \\ k \neq a_1, \dots, a_m}}^{N+m} C(\lambda_k) \end{aligned} \quad (2.106)$$

where the function G can be written as

$$G_{a_1, \dots, a_m}(\lambda_1, \dots, \lambda_{N+m}) = \prod_{j=1}^m d(\lambda_{a_j}) \frac{\prod_{\substack{b=1 \\ b \neq a_1, \dots, a_{j-1}}}^{N+j-1} \sinh(\lambda_{a_j} - \lambda_b + \eta)}{\prod_{\substack{b=1 \\ b \neq a_1, \dots, a_j}}^{N+j} \sinh(\lambda_{a_j} - \lambda_b)} \quad (2.107)$$

To proceed our calculation, we have to calculate the following "normalized" product:

$$\mathcal{S}(\{\lambda_1, \dots, \lambda_{N-m}, \xi_1, \dots, \xi_m\}, \{\lambda\}) = \frac{\langle 0 | \prod_{b=1}^{N-m} C(\lambda_b) \prod_{k=1}^m C(\xi_k) \prod_{k=1}^N B(\lambda_k) | 0 \rangle}{\langle 0 | \prod_{k=1}^N C(\lambda_k) \prod_{k=1}^N B(\lambda_k) | 0 \rangle} \quad (2.108)$$

By relation (2.93) and Gaudin formulas (2.95), we obtain:

$$\begin{aligned} &\mathcal{S}(\{\lambda_1, \dots, \lambda_{N-m}, \xi_1, \dots, \xi_m\}, \{\lambda\}) \\ &= \prod_{j,k=1}^m \frac{\sinh(\lambda_{N-m+k} - \lambda_{N-m+j})}{\sinh(\xi_k - \xi_j)} \prod_{j=1}^m \prod_{k=1}^{N-m} \frac{\sinh(\lambda_k - \lambda_{N-m+j})}{\sinh(\lambda_k - \xi_j)} \\ &\times \prod_{a=1}^N \prod_{k=1}^m \frac{\sinh(\lambda_a - \xi_k + \eta)}{\sinh(\lambda_a - \lambda_{N-m+k} + \eta)} \frac{\det \Psi'(\{\lambda\}, \{\xi\})}{\det \Phi'(\{\lambda\})} \end{aligned} \quad (2.109)$$

where Φ' is the Gaudin matrix. And for Ψ' we have

$$\begin{aligned} \Psi'_{ab} &= \Phi'_{ab} & b \leq N-m \\ \Psi'_{ab} &= \frac{\sinh \eta}{\sinh(\lambda_a - \xi_{b+m-N}) \sinh(\lambda_a - \xi_{b+m-N} + \eta)} & b > N-m \end{aligned} \quad (2.110)$$

As The Gaudin matrix is invertible, we get for the determinants

$$\frac{\det \Psi'(\{\lambda\}, \{\xi\})}{\det \Phi'(\{\lambda\})} = \det \left(\Phi'^{-1}(\{\lambda\}) \Psi'(\{\lambda\}, \{\xi\}) \right) \quad (2.111)$$

The first $N - m$ columns of the matrix $\Phi'^{-1}\Psi'$ are the unit matrix columns:

$$(\Phi'^{-1}\Psi')_{ab} = \delta_{ab}, \quad b \leq N - m \quad (2.112)$$

Using the representation for the Gaudin matrix in the thermodynamic limit 2.6.4 and the Lieb equation (2.77), we obtain:

$$\frac{1}{M} \sum_{b=1}^N \Phi'_{ab} \frac{\tilde{\rho}(\lambda_b - \xi_k + \frac{\eta}{2})}{\tilde{\rho}_{tot}(\lambda_b)} = \frac{\sinh \eta}{\sinh(\lambda_a - \xi_k) \sinh(\lambda_a - \xi_k + \eta)} + O(M^{-\infty}) \quad (2.113)$$

where

$$\begin{cases} \tilde{\rho}(\lambda) = \rho(\lambda), & -1 < \Delta \leq 1, \\ \tilde{\rho}(\lambda) = i\rho(i\lambda), & \Delta > 1. \end{cases} \quad (2.114)$$

Thus we obtain

$$(\Phi'^{-1}\Psi')_{ab} = \frac{\tilde{\rho}(\lambda_a - \xi_{b+m-N} + \frac{\eta}{2})}{M\tilde{\rho}_{tot}(\lambda_a)} + O(M^{-\infty}), \quad b > N - m \quad (2.115)$$

Finally the determinants in equation (2.109) can be written in a very simple way in the thermodynamic limit:

$$\frac{\det \Psi'(\{\lambda\}, \{\xi\})}{\det \Phi'(\{\lambda\})} = \frac{1}{M^m} \prod_{a=1}^m \tilde{\rho}_{tot}^{-1}(\lambda_{N-m+a}) \det S(\{\lambda_{N-m+1}, \dots, \lambda_N\}, \{\xi\}) + O(M^{-\infty}) \quad (2.116)$$

where

$$S_{ab} = \tilde{\rho}\left(\lambda_{N-m+a} - \xi_b + \frac{\eta}{2}\right) \quad (2.117)$$

Chapter 3

Experiment and Computational realization

3.1 Reproduction of experimental article

The Magnetization in dependent of the magnetic field is shown by fig 3.1. It can be shown that the saturation field is about $1.72T(\mu_0 H_s)$. In the numerical program of TBA equaitons, I assign the parameter J and g by taking use of the relations (3.1) with $M_s = 5212emu/mol$ and $H_s = 1.72$. This diagram shows the phase transformation from TLL to FP.

$$M_s[emu/mol] = N_A \mu_B g \times 1000 \quad H_s = \frac{4J}{g\mu_B} \quad (3.1)$$

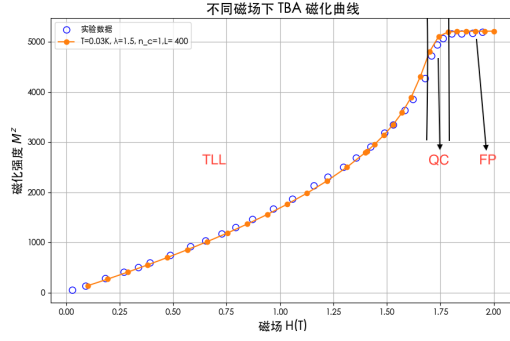


Figure 3.1

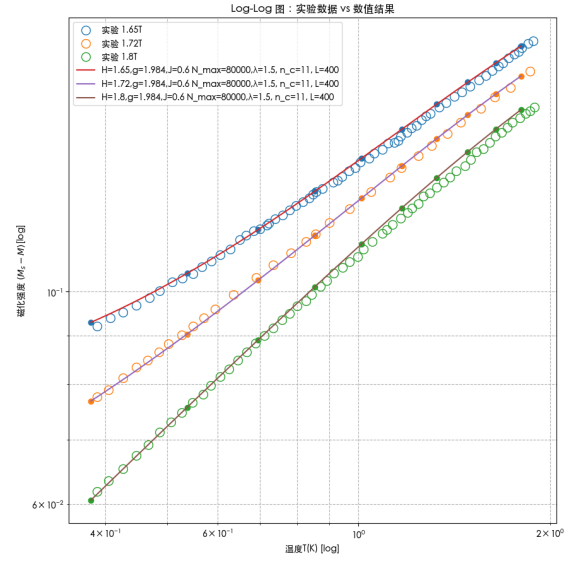


Figure 3.2

In the vicinity of saturation field H_s , it follows from (1.178) that the magnetization M is scaled by $T^{\frac{1}{2}}$ as in fig. 3.2. The linear behavior in the log-log diagram can be verified by noting

$$\begin{aligned} \frac{1}{2} - M &= T^{\frac{1}{2}} M(\Delta H/T) \\ \log\left(\frac{1}{2} - M\right) &= 1/2 \log(T) + \log(M(\Delta H/T)) \end{aligned} \quad (3.2)$$

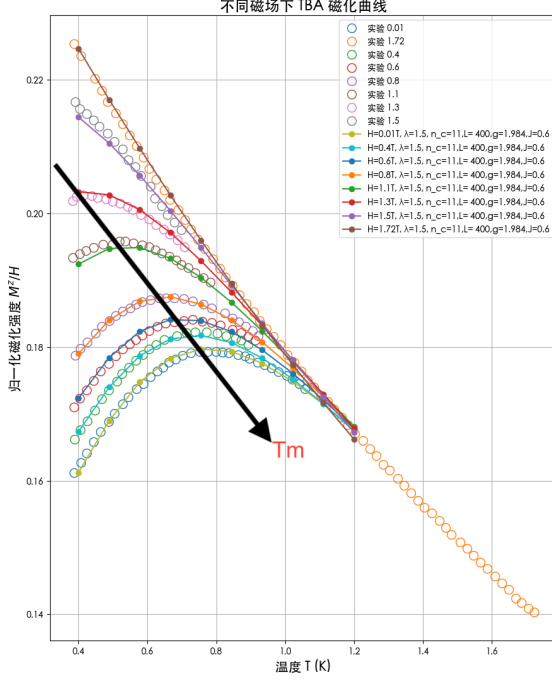


Figure 3.3

In the figure 3.3, the temperature T_m in the maximum can reflect a phase transition from TLL to QC. We can find that the line linking all maximum in the diagram bounded between TLL and QC. This property can be understood by noting that temperature depicted the width of critical region.

We can see from fig and fig the competition relation between quantum fluctuaion and thermal fluctuaion by noting that the peak magnetic field H_m in fig shifted to lower magnetic field with an increase in temperature and that the peak temperature T_m in fig shifted to lower temperature with an increase in magnetic field.

.1 Computational method

We start from the TBA equation:

$$\epsilon_n^+ = \epsilon_n^0 - \sum_{m=0}^{\infty} A_{mn} * \epsilon_n^- \quad (3)$$

where

$$\begin{aligned} \epsilon_n^+ &= T \ln[1 + \exp(\epsilon_n/T)] \quad ; \epsilon_n^- = -T \ln[1 + \exp(-\epsilon_n/T)] \\ \epsilon_n^0 &= -2\pi J a_n + nh = -\frac{nJ}{\lambda^2 + n^2/4} + nh \quad ; h = \mu_B g H \end{aligned} \quad (4)$$

Note that

$$\epsilon_n(\infty) = T \ln \left[\left(\frac{\sinh[(n+1)H/(2T)]}{\sinh[H/(2T)]} \right)^2 - 1 \right] \quad (5)$$

At $\lambda \rightarrow \infty$, (3) becomes:

$$\epsilon_n^+(\infty) = \epsilon_n^0(\infty) - \sum_{m=0}^{\infty} A_{mn} * \epsilon_n^+(\infty) \quad (6)$$

Substitute (6) from (3), we get:

$$\Delta\epsilon_n^+(\lambda) = -2\pi J a_n(\lambda) - \sum_{m=1}^{n_c} A_{mn} * \Delta\epsilon_n^-(\lambda) \quad (7)$$

Here we do truaction on n_c .

The equation for the dressed energy(the difference is at most a constant):

$$f = -\frac{1}{2}\epsilon_1^+(\infty) + \frac{h}{2} - \int_{-\Lambda}^{+\Lambda} G(\mu) \Delta\epsilon_1^+(\mu) d\mu \quad (8)$$

where

$$G(\lambda) = \frac{1}{2\cosh(\pi\lambda)}$$

We differentiate (8) over h and get:

$$\frac{\partial f}{\partial h} = -\frac{1}{2} \frac{\partial \epsilon_1^+(\infty)}{\partial h} + \frac{1}{2} - \int_{-\Lambda}^{+\Lambda} G(\mu) \frac{\partial \Delta\epsilon_1^+(\mu)}{\partial h} d\mu \quad (9)$$

where $\frac{\partial \epsilon_1^+(\infty)}{\partial h}$ is given by:

$$\frac{\partial \epsilon_n^+(\infty)}{\partial h} = (n+1) \coth h [(n+1)h/(2T)] - \coth h [h/(2T)] \quad (10)$$

We differentiate equation (??) over h and get:

$$\frac{\partial \Delta\epsilon_n^+(\lambda)}{\partial h} = - \sum_{m=1}^{n_c} A_{mn} * \frac{\partial \Delta\epsilon_n^-(\lambda)}{\partial h} \quad (11)$$

Some useful equations to do iterations on equation (7) are as below:

$$\begin{aligned} \epsilon_n(\lambda) &= \epsilon_n^+(\lambda) + \epsilon_n^-(\lambda) \\ \epsilon_n^-(\lambda) &= -T \ln \left(1 + e^{-\epsilon_n(\lambda)/T} \right) \end{aligned} \quad (12)$$

equivalently

$$\begin{aligned} \epsilon_n(\lambda) &= \Delta\epsilon_n^+(\lambda) + \Delta\epsilon_n^-(\lambda) = \epsilon_n(\infty) \\ \Delta\epsilon_n^-(\lambda) &= -T \ln \left(1 + e^{-\epsilon_n(\lambda)/T} \right) - \epsilon_n^-(\infty) \end{aligned} \quad (13)$$

The differential form is:

$$\frac{\partial \Delta\epsilon_n^-(\lambda)}{\partial h} = \frac{1}{e^{\epsilon_n(\lambda)/T} + 1} \left(\frac{\partial \Delta\epsilon_n^+(\lambda)}{\partial h} + \frac{\partial \Delta\epsilon_n^-(\lambda)}{\partial h} \right) - \frac{1}{e^{-\epsilon_n(\lambda)/T} + 1} \frac{\partial \epsilon_n(\infty)}{\partial h} + \frac{\partial \epsilon_n^+(\infty)}{\partial h} \quad (14)$$

where

$$\frac{\partial \epsilon_n(\infty)}{\partial h} = \frac{\coth\left(\frac{H}{2T}\right) \left(\cosh\left(\frac{H(n+1)}{T}\right) - 1 \right) - (n+1) \sinh\left(\frac{H(n+1)}{T}\right)}{\cosh\left(\frac{H}{T}\right) - \cosh\left(\frac{H(n+1)}{T}\right)}$$

Thus we get the expression for magnetization:

$$M_z = g\mu_B \frac{\partial f}{\partial h} \tag{15}$$